

Computational Method in Chemical Engineering T2730C2

Textbooks:

- Lorens T. Biegler, E. Ignacio Grossmann, Arthur W. Westerberg, Systematic Methods of Chemical Process Design, Prentice Hall International.
- Warren D. Seider, J. D. Seader, Daniel R. Lewin, Product and Process Design Principles: Synthesis, Analysis, and Evaluation, 2nd Edition, Wiley.
 - Robin Smith, Chemical Process: Design and Integration, Wiley.
- James M. Douglas, Conceptual Design of Chemical Processes, McGraw Hill International, 1988.

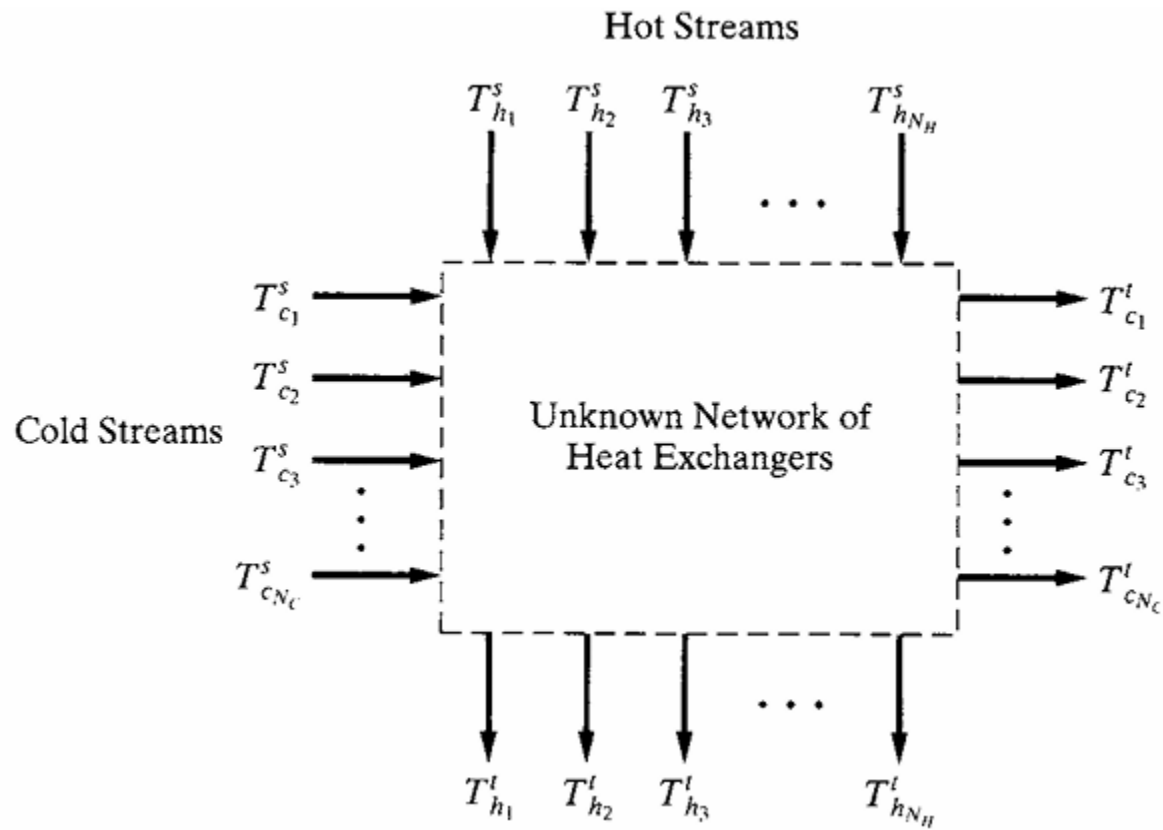
Unit 4 – Synthesis of Heat Exchanger Networks

- Basic Heat Exchanger Network Synthesis (HENS)
- Minimum Utility Targets
- Temperature Interval Method
- Hohmann / Lochart Composite Curves (HCC)
- Grand Composite Curves (GCC)
- Pinch Design Approach to Inventing a Network
- Networks for Maximum Energy Recovery
- Minimum Number of Exchangers
- Stream Splitting
- Threshold and Optimum Approach Temperature
- Derivation of Network Superstructures for Minimization of Annual Costs
- Multiple Utility Design Problems

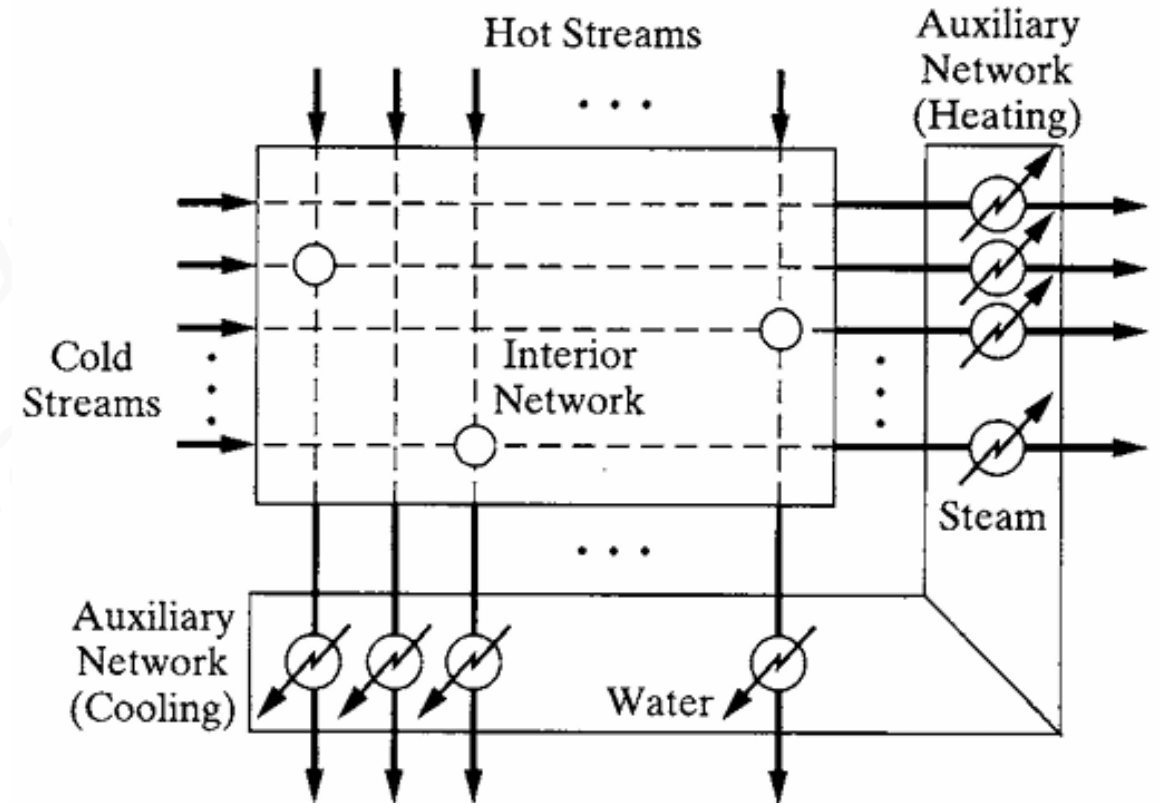
Basic Heat Exchanger Network Synthesis (HENS)

- Heat and power integration seeks to utilize the energy in the high-temperature streams that need to be cooled and/or condensed to heat and/or vaporize the cold streams, and provide power to compressors from turbines and heat engines where possible.
- In most designs, it is common initially to disregard power demands in favor of **designing an effective network of heat exchangers by heat integration**, without using the energy of the high-temperature streams to produce power.
- To accomplish this, N_H hot process streams, with specified source and target temperatures T_{hi}^s and T_{hi}^t are cooled by N_c cold process streams, with specified source and target temperatures, T_{cj}^s and T_{cj}^t .
- It is common to refer to the **heat exchangers between the hot and cold process streams as comprising the interior network**, and those between the hot or cold streams and the utilities as comprising the auxiliary network.

Heat integration schematics



(a) source and target temperatures for heat integration



(b) interior and auxiliary networks of heat exchangers

Basic Heat Exchanger Network Synthesis (HENS)

- It is desired to synthesize the **most economical network of heat exchangers**, and when generating and comparing alternative flowsheets, an **approximate profitability measure** is sufficient, such as the **annualized cost**.

$$C_A = i_m (C_{TCI}) + C$$

- where C_{TCI} is the total capital investment, i_m is a reasonable return on investment annually, and C is the annual cost of sales, as defined in the *cost sheet*.
- Approximate C_{TCI} as the **sum of the purchase costs for each of the heat exchangers** (without including installation costs and other capital investment costs).
- The **purchase costs can be estimated based on the area for heat transfer A** , estimated from the heat transfer rate equation.

$$A = Q/(UF_T\Delta T_{LM})$$

- where Q is heat duty, U is overall heat-transfer coefficient, F_T is correction factor for a multiple-pass exchanger (0.75-0.9 or optimal = 1.0), and ΔT_{LM} is log-mean temperature-driving force for countercurrent flow.

Basic Heat Exchanger Network Synthesis (HENS)

- Approximate **C** as the annual cost of the utilities for heating and cooling, typically using steam and cooling water.

$$C_A = i_m \left[\sum_i C_{P,I_i} + \sum_j C_{P,A_j} \right] + sF_s + (cw)F_{cw}$$

- Where C_{P,I_i} and C_{P,A_j} are the purchase costs of the heat exchangers in the interior and auxiliary networks, F_s is the annual flow rate of steam (kg/year), s is the unit cost of steam (rupees/kg), F_{cw} is the annual flow rate of cooling water, and cw is the unit cost of cooling water.
- When other utilities, such as fuel, cool air, boiler feed water, and refrigerants are used, additional terms are needed.
- Approaches have been developed to optimize:**
 - A network of heat exchangers is designed having the **minimum usage of utilities (MER network)**, usually requiring a large number of heat exchangers.
 - The **number of heat exchangers is reduced toward the minimum**, possibly at the expense of increasing the consumption of utilities.

Basic Heat Exchanger Network Synthesis (HENS)

- If step 2 is implemented and no. of HE reduced one by one, then capital costs are reduced due to economy of scale by $C_p = KA^n$ ($n < 1$, 0.6).
- As no. of HEs reduced, total area for HT constant, area of each HE increases, as $n < 1$ purchase cost per unit of area is decreased.
- However, the **consumption of utilities is normally increased**.
- At some point, the increased cost of utilities overrides the decreased cost of capital and CA increases beyond the minimum.
- When the cost of fuel is high, the minimum C_A is not far from that for a network of heat exchangers using the minimum utilities.
- ***The selection of the minimum approach temperature, ΔT_{min} for the heat exchangers is a key design variable in the synthesis of heat exchanger networks (HENs), because of its impact on lost work associated with heat transfer.***

Basic Heat Exchanger Network Synthesis (HENS)

- For the heat transfer between the high and low-temperature reservoirs, where LW is the rate of lost work and T_0 is the absolute temperature of the environment.

$$\begin{aligned} LW &= \left(1 - \frac{T_0}{T_1}\right)Q + \left(1 - \frac{T_0}{T_2}\right)(-Q) &&= Q\left(\frac{T_0}{T_2} - \frac{T_0}{T_1}\right) \\ &= QT_0\left(\frac{T_1 - T_2}{T_1T_2}\right) &&= QT_0\frac{\Delta T}{T_1T_2} \end{aligned}$$

- For a given rate of heat transfer and ΔT approach, the rate of lost work increases almost inversely with the decrease in the square of the absolute temperature level.**
- If the approach temperature differences were not reduced, the large increases in the rate of lost work would sharply increase the energy requirements to operate these processes, especially the operating and installation costs for compressors.

Minimum Utility Targets

- A principal objective in the synthesis of HENs is the efficient utilization of energy in the hot process streams to heat cold process streams.
- Thus, it is desirable to compute the ***maximum energy recovery (MER) before synthesizing the HEN***; and to determine the minimum hot and cold utilities in the network, given the heating and cooling requirements of the process streams.
- This important first step is referred to as *MER targeting*, and is useful in that it determines the utility requirements for the most thermodynamically efficient network.
- To introduce this targeting step, the example provided by Linnhoff and Turner (1981) is analyzed, however it involves only sensible heat.

Minimum Utility Targets – Example 10.1

Two cold streams, C1 and C2, are to be heated and two hot streams, H1 and H2, are to be cooled without phase change. Their conditions and properties are as follows:

Stream	T^s (°F)	T^t (°F)	mC_p [Btu/(hr°F)]	$Q(10^4 \text{ Btu/hr})$
C1	120	235	20,000	230
C2	180	240	40,000	240
H1	260	160	30,000	300
H2	250	130	15,000	180

- It is assumed that the heat-capacity flow rate, $mC_p = C$, which is the product of the specific heat and the mass flow rate, does not vary with temperature.
- **Design a HEN that uses the smallest amounts of heating and cooling utilities possible, such that the closest approach temperature differences never fall below a minimum value.**
- For the temperature range in this example, a reasonable assumption is $\Delta T_{\min} = 10^\circ\text{F}$.

Minimum Utility Targets – Example 10.1

- For this system, a total of 480×10^4 Btu/hr must be removed from the two hot streams, but only a total of 470×10^4 Btu/hr can be consumed by the two cold streams.
- Hence, from the first law of thermodynamics, a **minimum of 10×10^4 Btu/hr must be removed by a cold utility such as cooling water.**
- This is not the minimum utility usage, which, from the second law of thermodynamics, depends on $\Delta T_{\min}$.
- **Three methods are introduced to estimate MER targets:**
 - (1) the temperature-interval method
 - (2) a graphical method using composite heating and cooling curves
 - (3) the formulation and solution of a linear programming (LP) problem.

Temperature Interval Method

- The temperature-interval method was developed by Linnhoff and Flower (1978a, b) following the pioneering work of Hohmann (1971).
- The method is applied to the hot and cold streams, and a systematic procedure unfolds for determining the minimum utility requirements over all possible HENs, given just the heating and cooling requirements for the process streams and the minimum approach temperature in the heat exchangers, $\Delta T_{\min}$.
- **Check manual experiment 3 for solution of Example 10.2.**
- **Video: <https://www.youtube.com/watch?v=7PpysQMD0WE>**

Hohmann / Lochart Composite Curves (HCC)

- The terminology, pinch, is understood more clearly in connection with a graphical display, introduced by Umeda et al. (1978), in which composite heating and cooling curves are positioned no closer than $\Delta T_{\min}$.
- **Composite curves are T/H diagrams (temperature/heat diagrams) used to visualize cold and hot streams and potential heat transfer between them.**
- It is a graphical technique used for visualizing the heat cascade.

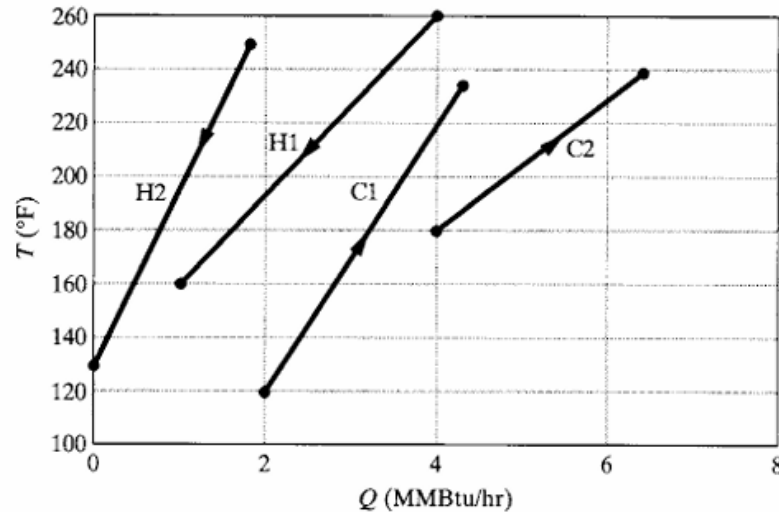
$$Q = \int_{T_s}^{T_t} C_p dT = C_p (T_t - T_s) = \Delta H$$

$$\frac{dT}{dQ} = \frac{1}{C_p}$$

- The slope of a composite curve is the inverse of the heat-capacity flow rate, C , as C is constant and not a function of T , the curves are straight lines.
- For the hot streams, they are cooling curves that begin at the highest temperature and finish at the lowest temperature after the energy has been removed. For the cold streams, they are heating curves that begin at the lowest temperature and finish at the highest temperature after heat has been added.

Hohmann / Lochart Composite Curves (HCC)

Heating and cooling curves



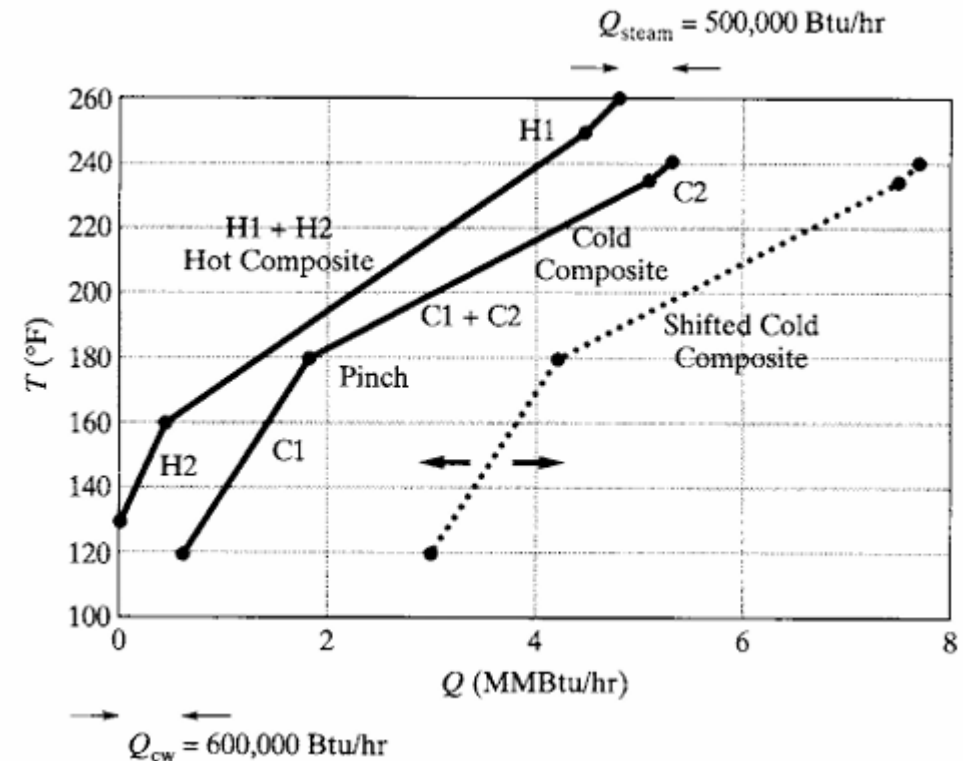
Hot composite enthalpies

T (°F)	130	160	190	245	250	260
H (Btu/hr $\times 10^{-4}$)	0	45	180	427.5	450	480

Cold composite enthalpies

T (°F)	120	150	180	235	240
H (Btu/hr $\times 10^{-4}$)	60	120	180	510	530

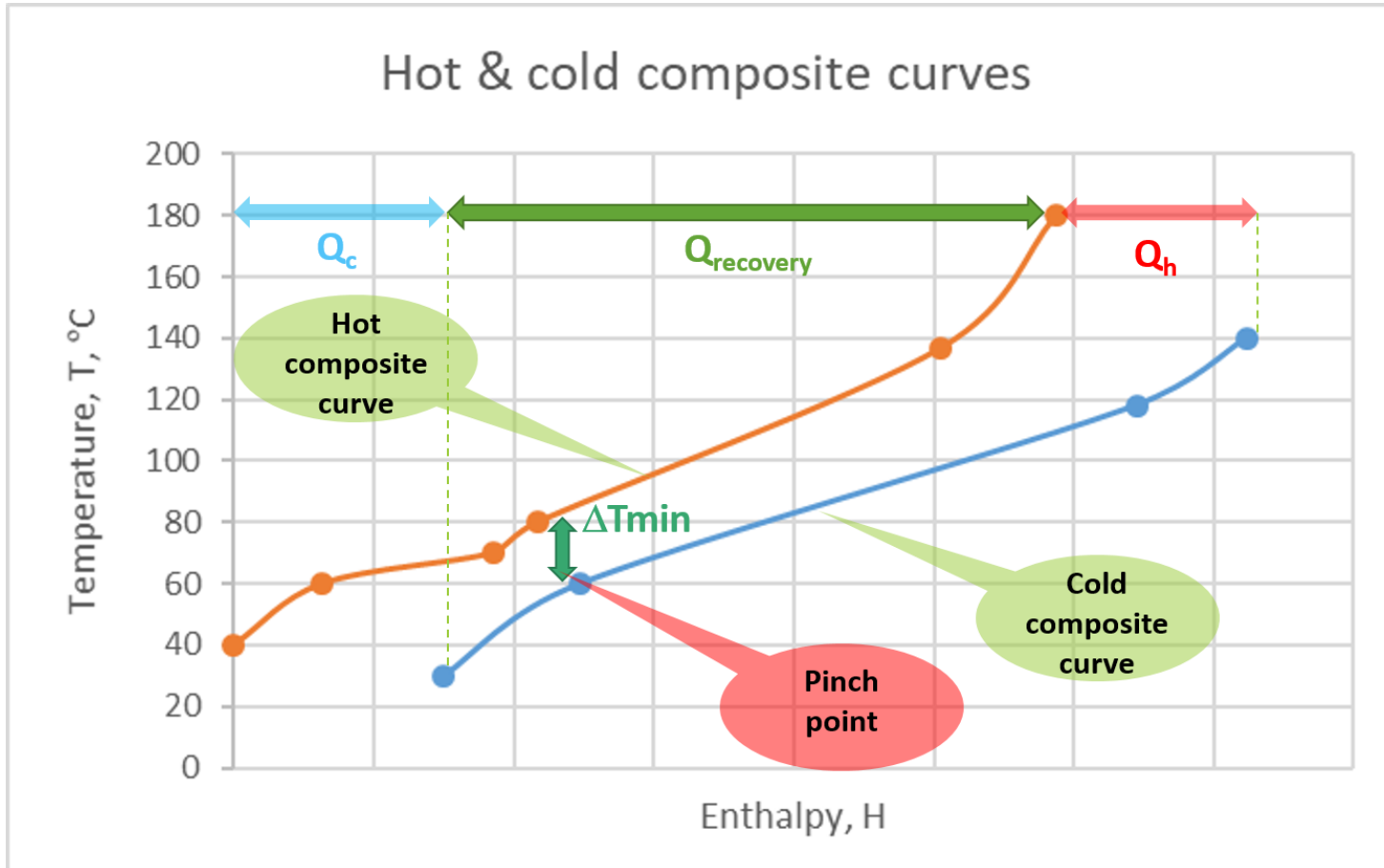
Hohmann / Lochart Composite Curves (HCC)



Hohmann / Lochart Composite Curves (HCC)

- The composite curves have a closest point of approach of $\Delta T_{\min} = 10^{\circ}\text{F}$ at the point where stream C2 begins along the cold composite curve; 180°F . The corresponding temperature on the hot composite curve is 190°F .
- Consequently, these two points provide the temperatures at the pinch.
- *If $\Delta T_{\min}$ is reduced to zero, the cold composite curve is shifted to the left until it touches the hot composite curve, this corresponds to an infinite area for heat exchange.*
- **The $\Delta T_{\min}$ at which the pinch disappears is referred to as the *threshold* $\Delta T_{\min}$.**
- At the **high-temperature ends**, however, no segments of the hot composite curve lie vertically above the upper end of the cold composite curve.
- There, **an additional 50×10^4 Btu/hr must be supplied from a hot utility, such as steam.**
- Similarly, at the **low-temperature ends of the composite curves**, no segments of the cold composite curve lie vertically below the lower end of the hot composite curve. An additional **60×10^4 Btu/hr must be removed using a cold utility, such as cooling water.**

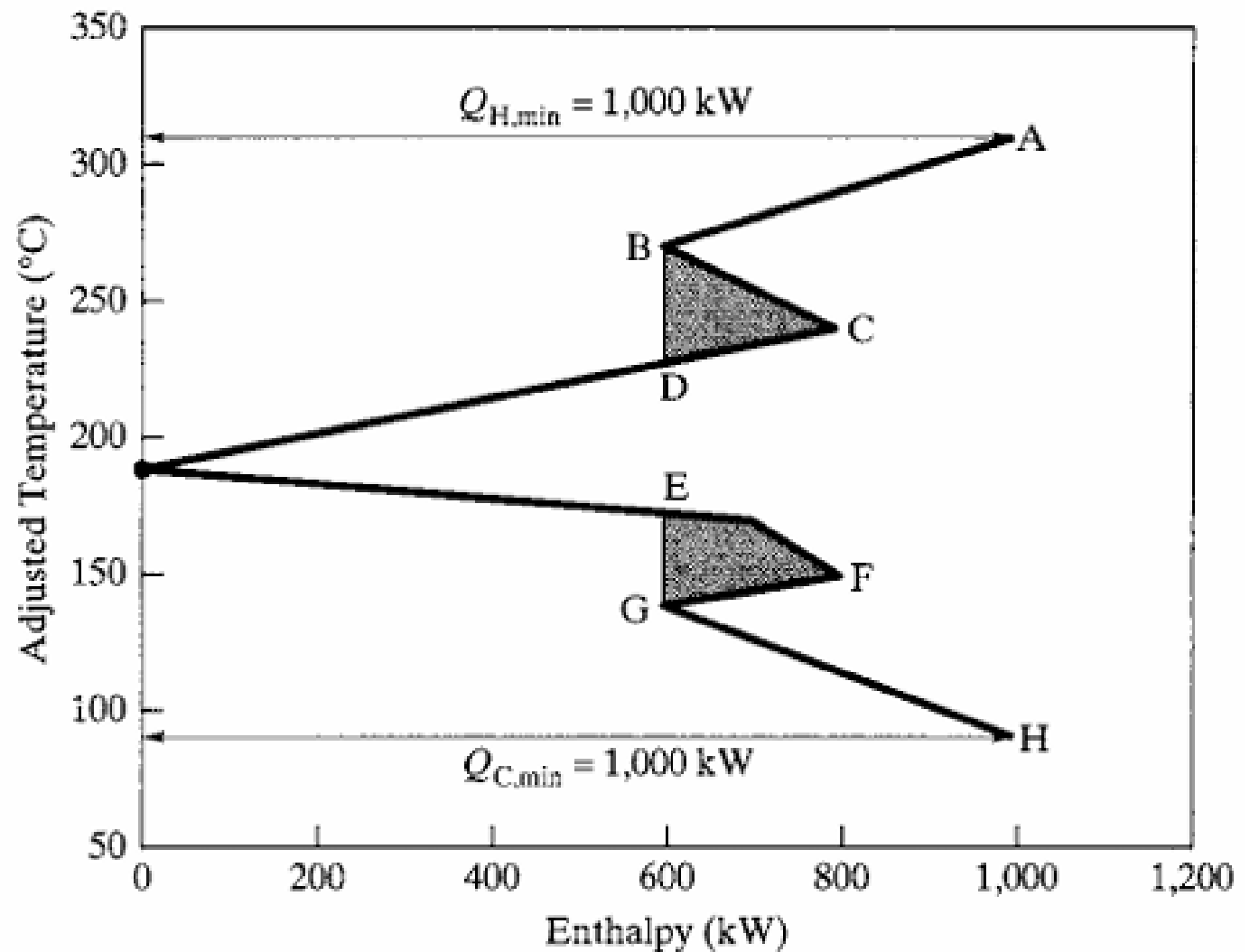
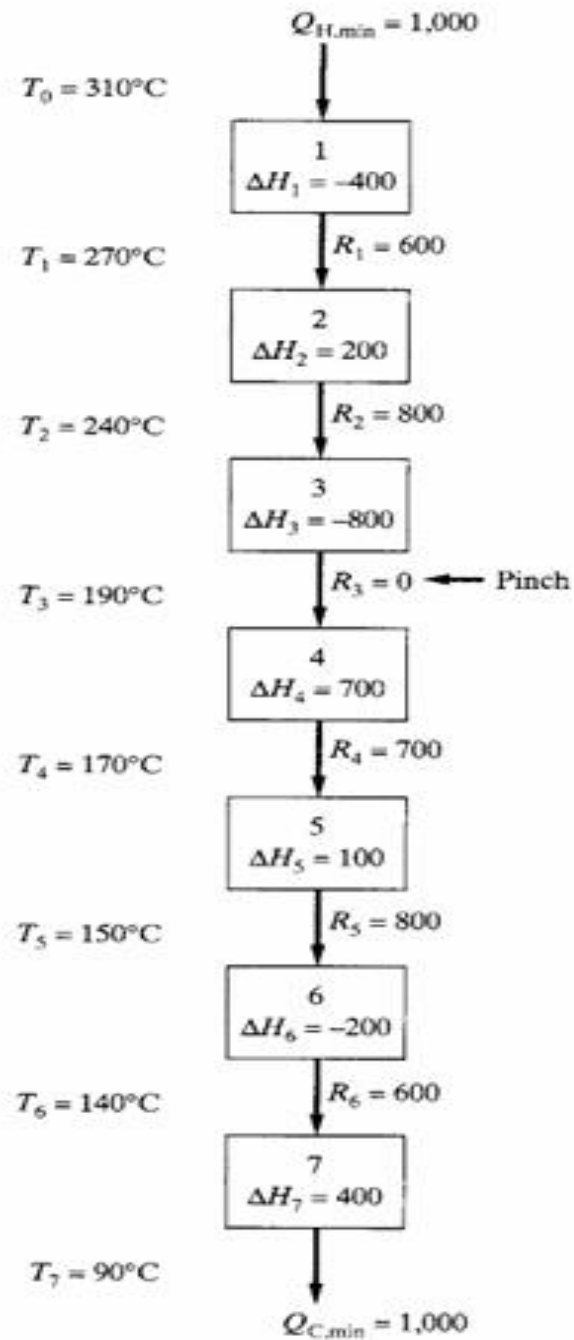
Hohmann / Lochart Composite Curves (HCC)



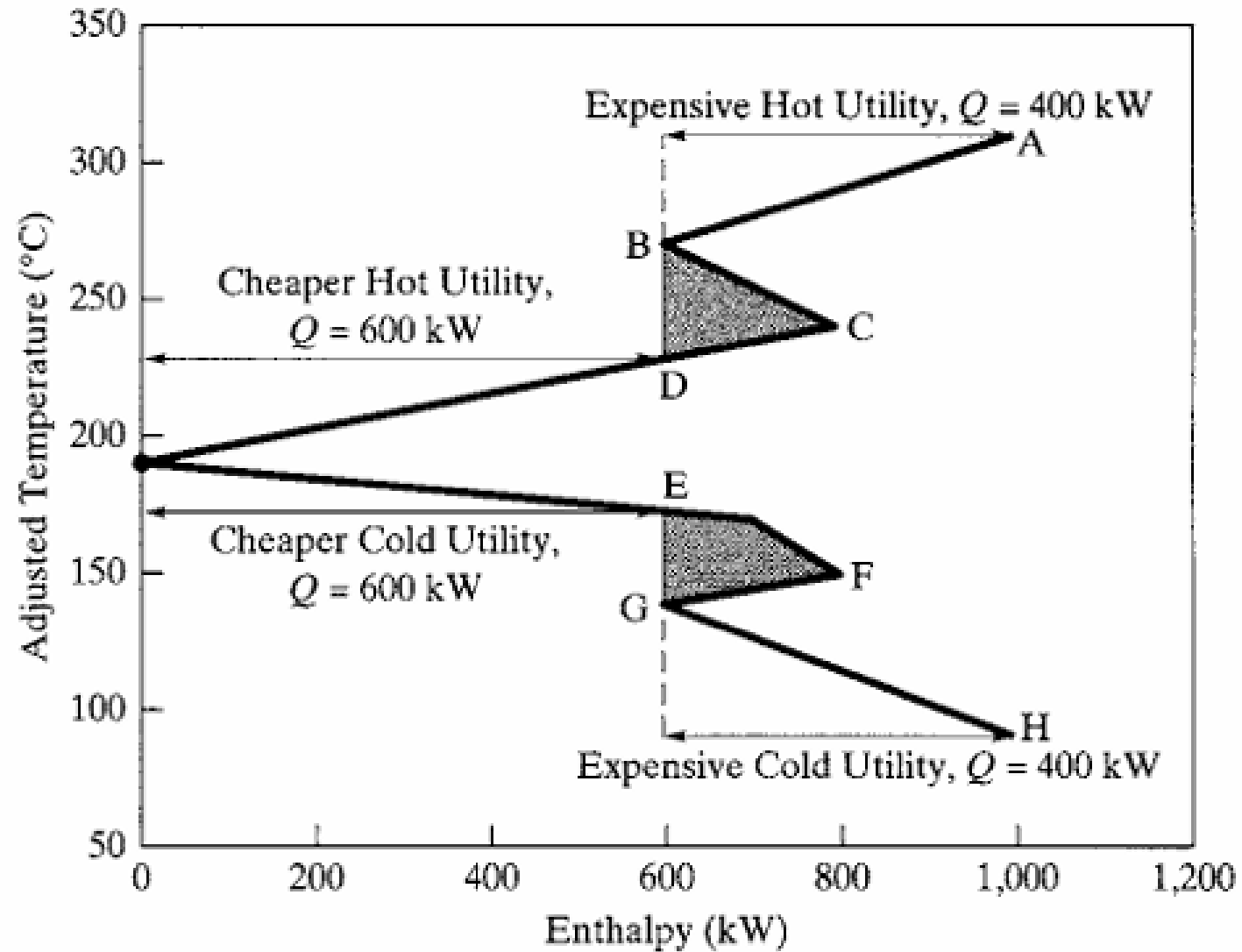
- The overlap between the composite curves represents the maximum amount of heat recovery possible within the process.
- The “overshoot” at the bottom of the hot composite represents the minimum amount of external cooling required.
- The “overshoot” at the top of the cold composite represents the minimum amount of external heating required.
- The position where two composite curves are the closest to each other represents the pinch point that will define the requirements of how to improve energy efficiency.

Grand Composite Curves (GCC)

- In the grand composite curve (GCC), the enthalpy residuals are displayed as a function of the interval temperatures.
- The enthalpy residuals corresponding to the highest and lowest interval temperatures are the minimum heating and cooling utility duties.
- The local rise in the enthalpy residual along the arc BC, which indicates an increase in the residual heat available, can be used to supply the increased demand denoted by the decrease in the enthalpy residual along the arc CD. This is accomplished using **internal heat exchange**.
- Similarly, the heat demand along the arc FG can be supplied by internal heat exchange with the arc EF.
- When 1,000 kW are provided at 330°C, a fired-heater is required that burns fuel.



Grand composite curve for temperature intervals, energy balances, and residuals determined



Grand composite curve for the temperature intervals showing possible savings by utilizing HPS and BFW.

Grand Composite Curves (GCC)

- Alternatively, the GCC replotted in Figure shows that up to 600 kW can be provided at 230°C using less expensive high-pressure steam at 400 psi, noting that the alternative utility temperature level and duty are limited by the intercept D.
- This leaves only 400 kW to be supplied in the fired-heater.
- Similarly, instead of removing 1,000 kW at 90°C using cooling water, up to 600 kW can be removed at 170°C by heating boiler feed water, leading to further savings in operating costs.
- As before, the temperature level and duty are limited by intercept E.
- Evidently, **significant savings in operating costs are revealed by the GCC.**
- **The grand composite curve helps to better utilize utility resources when designing HENs.**

MER targeting for a process exhibiting phase changes and variable heat capacities

- Example: for the ternary reactor effluent, only vapor sensible heat is involved from 800°F to 183°F (dew point). Then, condensation occurs, involving both latent heat and sensible heat to the target temperature of 65°F.
- **Piece-wise linear approximations, defined by critical coordinates on the original heating and cooling curves.**
- The piece-wise linear approximations are defined in terms of temperature-enthalpy coordinates, (h_k, T_k) and (h_{k+1}, T_{k+1}) through which linear arcs are drawn to approximate the true heating or cooling curves.

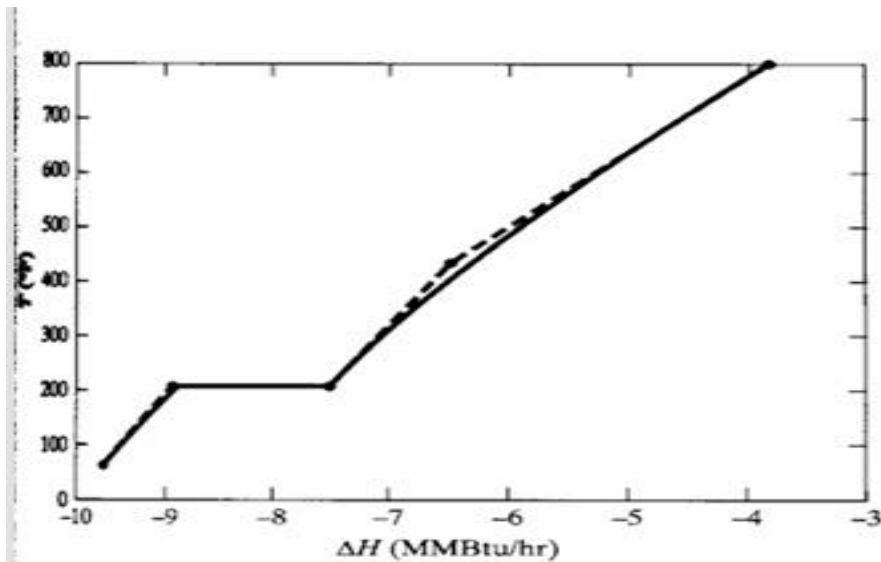


Figure 10.9 Temperature-enthalpy diagram for the cold stream, S2, showing simulation results (solid line) and piece-wise linear approximation (dashed line).

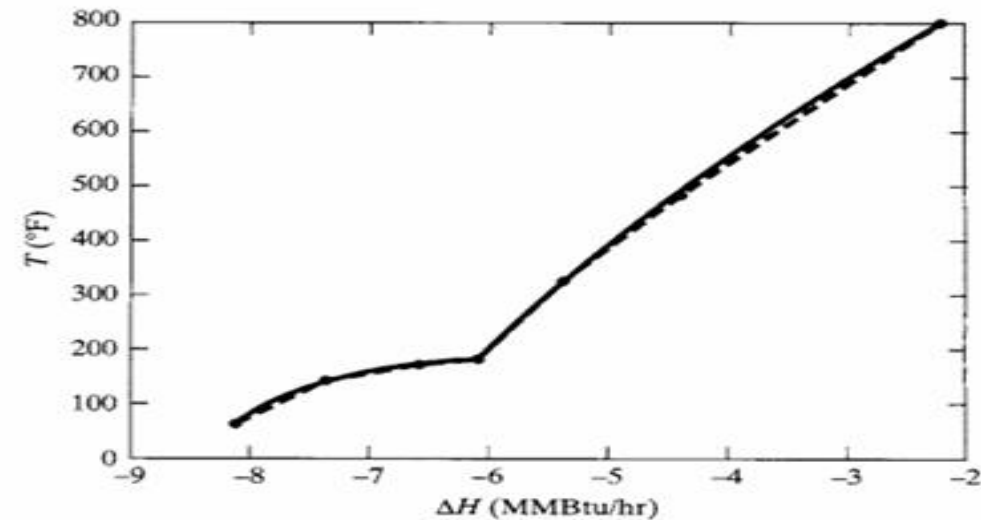


Figure 10.10 Temperature-enthalpy diagram for the hot stream, S4, showing simulation results (solid line) and piece-wise linear approximation (dashed line).

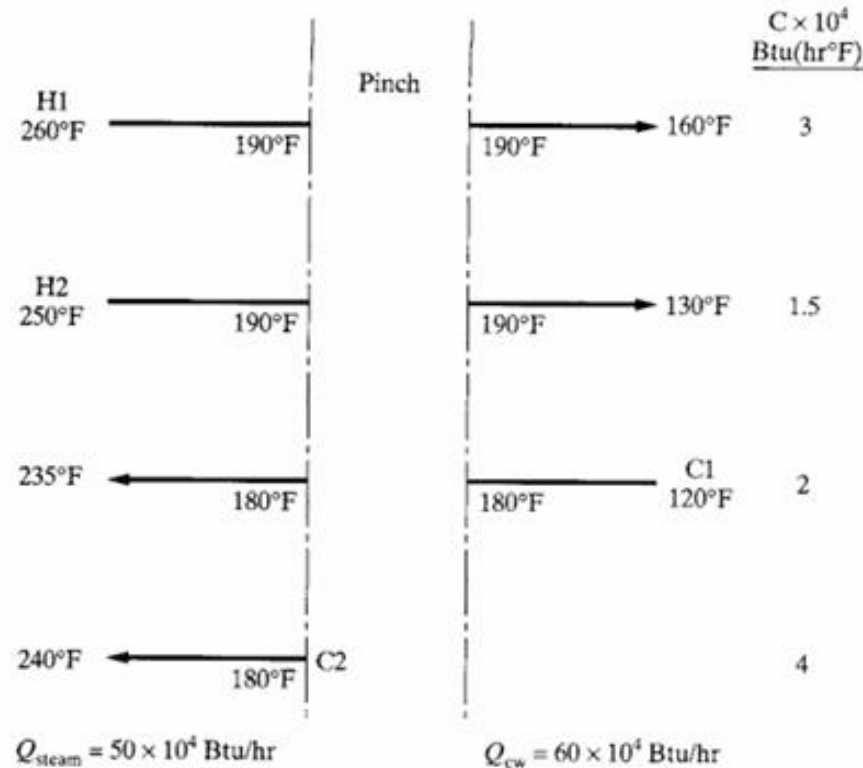
Networks for Maximum Energy Recovery

- **Having determined the minimum utilities for heating and cooling, it is common to design two networks of heat exchangers, one on the hot side and one on the cold side of the pinch.**

Two methods :

1. Introduced by Linnhoff and Hindmarsh (1983), places emphasis on positioning the heat exchangers by working out from the pinch.
2. The second is an algorithmic strategy that utilizes a mixed-integer linear program (MILP), which was introduced by Papoulias and Grossmann (1983b) and is solved with GAMS.

Stream Matching at the Pinch



- Attention is focused at the pinch where the temperatures of the hot and cold streams are separated by $\Delta T_{\min}$.
- This, of course, is the location of the closest temperature approach.
- For a counter current heat exchanger, the temperature approach on cold side is ΔT_1 and the temperature approach on hot side is ΔT_2 .

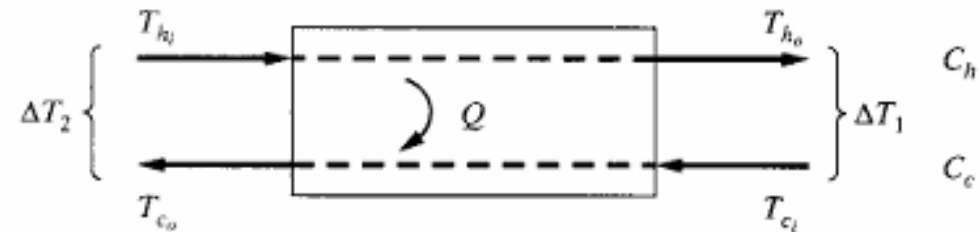
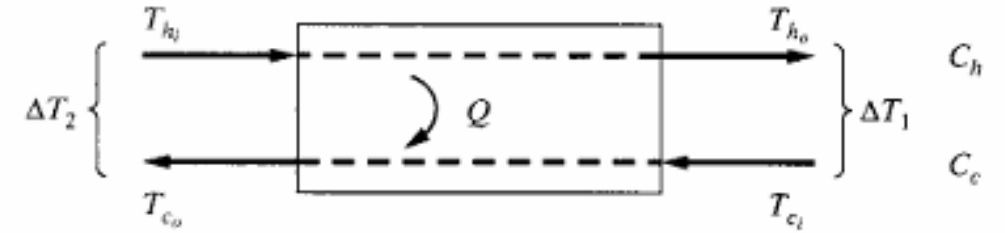


Figure. Pinch decomposition of hot and cold streams

Stream Matching at the Pinch



- Carrying out energy balance for cold and hot streams

$$Q = C_h (T_{h_i} - T_{h_o}) \quad \text{or} \quad T_{h_i} - T_{h_o} = \frac{Q}{C_h}$$

$$Q = C_c (T_{c_o} - T_{c_i}) \quad \text{or} \quad T_{c_o} - T_{c_i} = \frac{Q}{C_c}$$

Subtracting eq 2 from 1

$$(T_{h_i} - T_{c_o}) - (T_{h_o} - T_{c_i}) = Q \left(\frac{1}{C_h} - \frac{1}{C_c} \right)$$

$$\Delta T_2 - \Delta T_1 = \frac{Q(C_c - C_h)}{C_h C_c}$$

Stream Matching at the Pinch

- Following the approach introduced by Linnhoff and Hindmarsh (1983), the potential locations for the heat exchangers at the pinch are considered next.
- **When a heat exchanger is positioned on the hot side of the pinch, which is considered first arbitrarily, $\Delta T_1 = \Delta T_{\min}$.**

$$\Delta T_2 = \Delta T_{\min} + \frac{Q(C_c - C_h)}{C_h C_c}$$

- To assure that $\Delta T_2 > \Delta T_{\min}$ since $Q > 0$ and the heat-capacity flow rates are positive, it follows that $C_c \geq C_h$ is a necessary and sufficient condition.
- **For a match to be feasible at the pinch, on the hot side, $C_c \geq C_h$ must be satisfied.**
- If two streams are matched at the pinch with $C_c < C_h$, the heat exchanger is infeasible because $\Delta T_2 < \Delta T_{\min}$.

Stream Matching at the Pinch

- When a heat exchanger is positioned on the cold side of the pinch, $\Delta T_2 = \Delta T_{\min}$.

$$\Delta T_1 = \Delta T_{\min} - \frac{Q(C_c - C_h)}{C_h C_c}$$

- In this case, to assure that there are no approach temperature violations ($\Delta T_1 \geq \Delta T_{\min}$), it is **necessary and sufficient that $C_h \geq C_c$** .
- This condition is just the reverse of that on the hot side of the pinch.

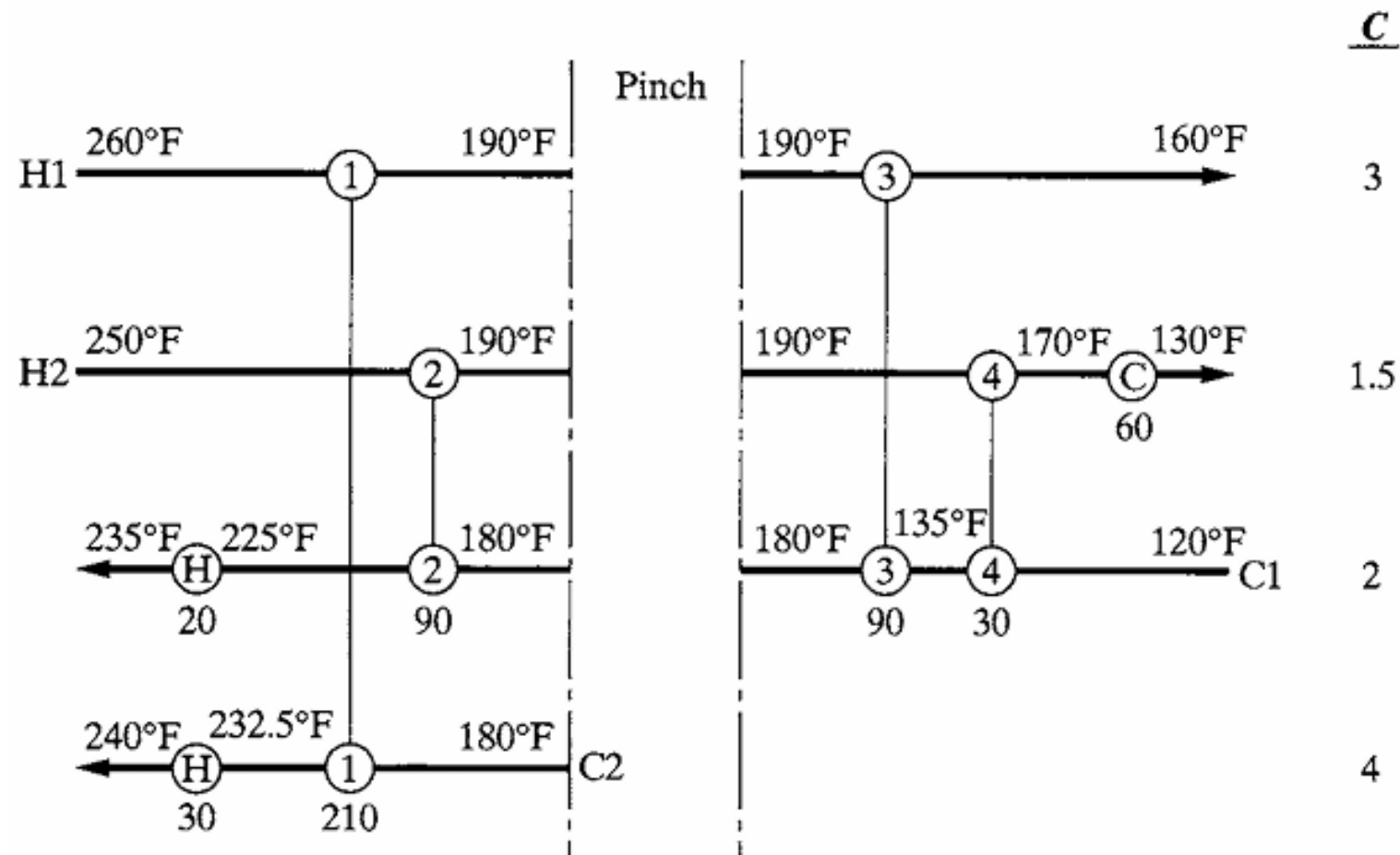
Steps to design HEN to meet MER targets

1. MER Targeting: The pinch temperatures are determined, together with minimum hot and cold utility targets, $Q_{H,min}$ and $Q_{C,min}$. Either the temperature-interval or composite curve methods are used, or a linear programming problem is formulated and solved.
2. The synthesis problem is decomposed at the pinch, yielding two independent HENs to be designed. It is helpful to place the heat-capacity flow rates for each stream in a column to the right, for reference.
3. The HEN is designed on the hot side of the pinch, starting at the pinch, and working outwards. At the pinch, streams are paired such that $C_c \geq C_h$. Hot utilities are added to meet cold temperature targets (up to a total of $Q_{H,min}$).
4. The HEN is designed on the cold side of the pinch, starting at the pinch, and working outwards. At the pinch, streams are paired such that $C_h \geq C_c$. Cold utilities are added to meet cold temperature targets (up to a total of $Q_{C,min}$).

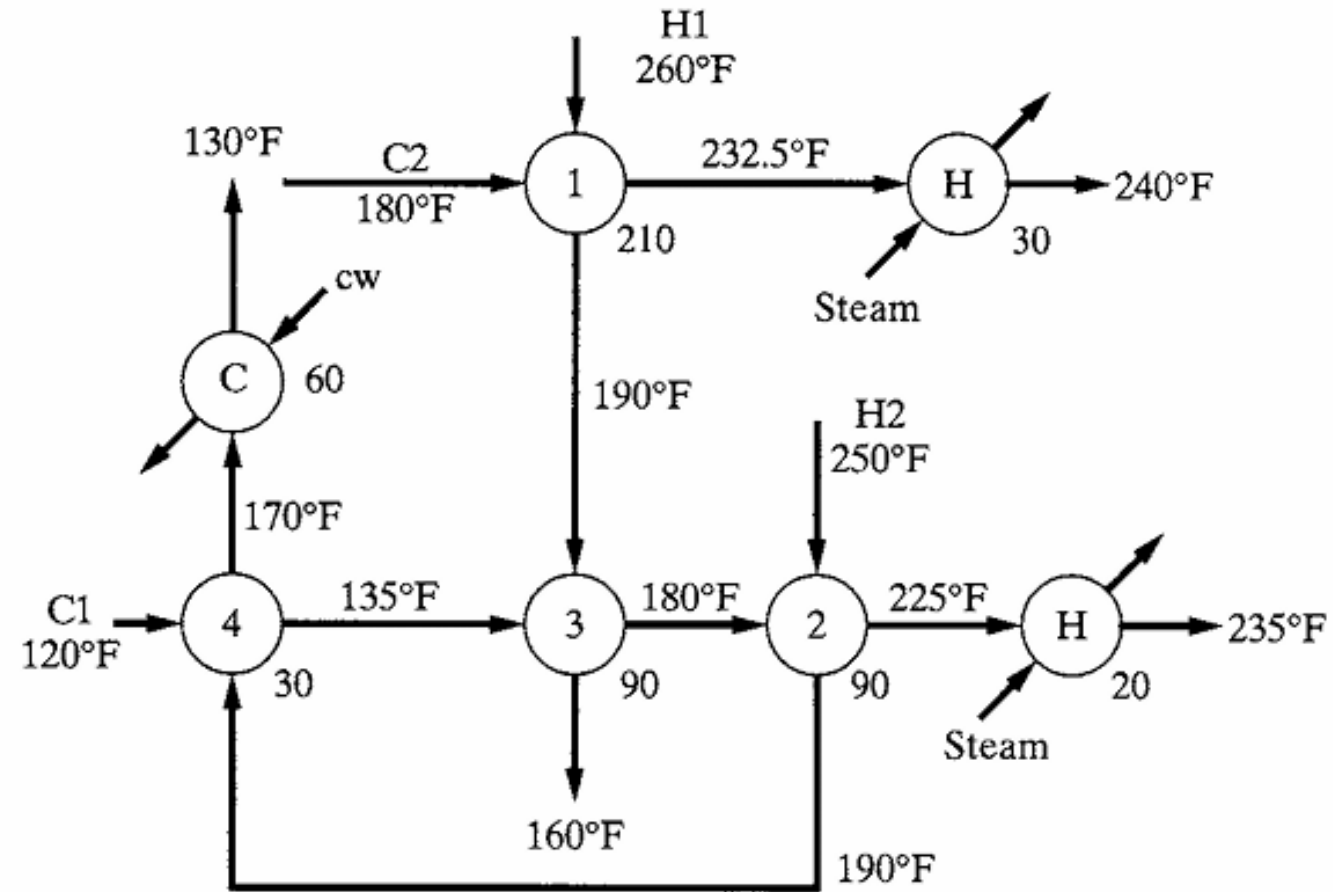
Example 10.6

- Design a HEN to meet the MER targets for Example 10.1: $Q_{Hmin} = 50 \times 10^4$ Btu/hr and $Q_{Cmin} = 60 \times 10^4$ Btu/hr, where Q_{Hmin} and Q_{Cmin} are the minimum hot and cold utility loads.
- When designing a HEN to meet MER targets, no heat is transferred across the pinch, and hence, two HENs are designed, one on the hot side and one on the cold side of the pinch.

Interior HEs and auxiliary HEs



Flowsheet for HEN



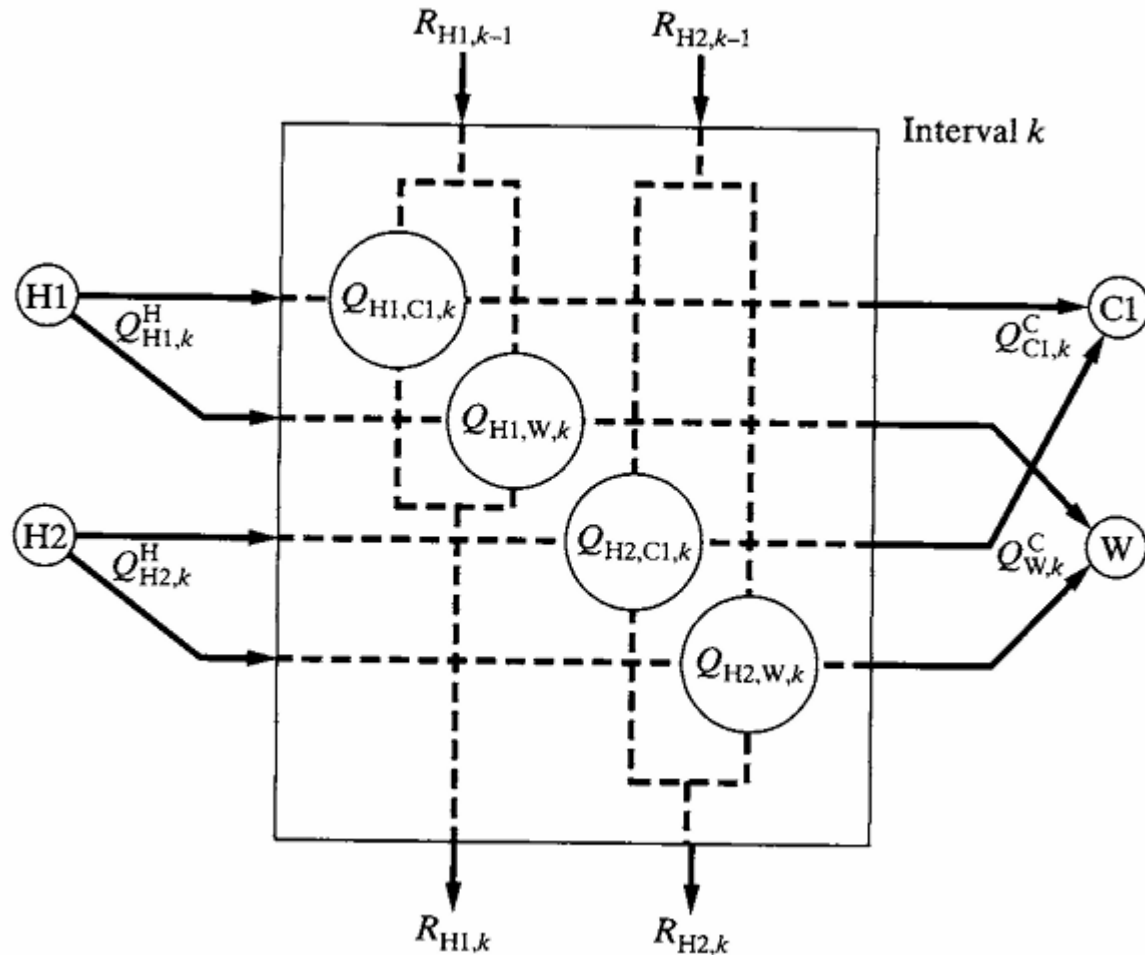
Mixed-Integer Linear Programming

- The **transshipment model for heat transfer**, introduced by Papoulias and Grossmann (1983b), provides a more systematic method for stream matching.
- In this model, the **hot streams and hot utilities are viewed as the sources of energy** that distribute among the temperature intervals, and that in turn become a source of energy for the **sinks (or destinations), that is, the cold streams and cold utilities**.
- The temperature levels of the sources and sinks establish the temperature intervals within which the sources can provide energy, and similarly, within which the sinks can receive energy.
- Energy can also be carried by the hot streams, as residuals, to the adjacent intervals at a lower temperature.
- **A superstructure in which all possible exchanges of heat between the hot streams, cold streams, and cold utility are included.**

Mixed-Integer Linear Programming

- **Using this superstructure to design a network with a low capital cost having the minimum utilities**, Papoulias and Grossmann created a MILP that minimizes the number of matches (i.e., that minimizes the number of heat exchangers).
- The **objective is to minimize the number of matches**, and hence, the objective function sums over all of the possible matches, with the weighting coefficient, increased as certain matches become less desirable.

Mixed-Integer Linear Programming



$$\begin{array}{ll} \text{Minimize } z = \sum_i \sum_j w_{ij} y_{ij} \\ \text{w.r.t} \end{array}$$

$$Q_{ijk}, y_{ij}$$

s.t.

$$R_{ik} - R_{i,k-1} + \sum_{j \in C_k} Q_{ijk} = Q_{ik}^H \quad i \in H_k \quad k = 1, \dots, K$$

$$\sum_{i \in H_k} Q_{ijk} = Q_{ik}^C \quad j \in C_k \quad k = 1, \dots, K$$

$$\sum_k Q_{ijk} - y_{ij} U_{ij} \leq 0 \quad i \in H \quad j \in C$$

$$R_{ik} \geq 0$$

$$R_{i0} = R_{iK} = 0$$

$$Q_{ijk} \geq 0 \quad y_{ij} \in \{0, 1\}$$

Minimum Number of Exchangers

- **Having designed a HEN that meets the MER targets, it is common to reduce the number of heat exchangers toward the minimum** while permitting the consumption of utilities to rise, particularly when **small heat exchangers can be eliminated**.
- In this way, lower annualized costs may be obtained, especially when the cost of fuel is low relative to the purchase cost of the heat exchangers.
- Alternatively, an attempt can be made to design the HEN to minimize the number of heat exchangers and satisfy the MER target, by invoking stream splitting.

Reducing the Number of Heat Exchangers-Breaking Heat Loops

- Hohmann (1971), the **minimum number of heat exchangers in a HEN** is

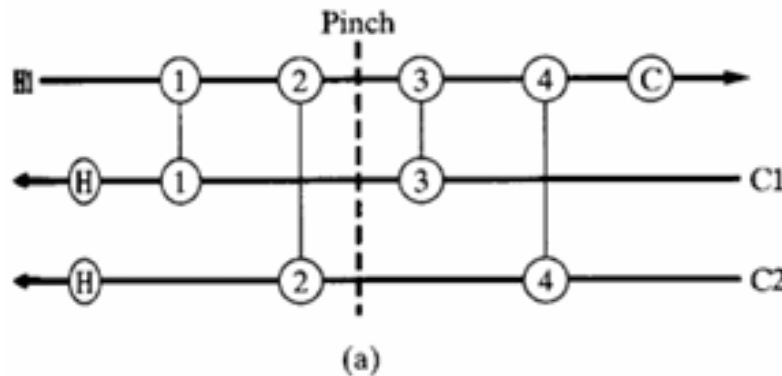
$$N_{HX,min} = N_S + N_U - 1$$

- where N_S is the number of streams and N_U is the total number of distinct hot and cold utility sources.
- Hot utilities: fuel, steam at high pressure (HPS), intermediate pressure (JPS), and low pressure (LPS).
- Cold utilities: boiler feed water (BFW), cooling water (CW), and refrigeration.
- Hence, for the networks in Examples 10.7, which involve four streams and two utilities (steam and cooling water), $N_{HX,min} = 4+2-1 = 5$.

Reducing the Number of Heat Exchangers- Breaking Heat Loops

- Douglas (1988) shows that the **minimum number of heat exchangers is also dependent on the number of *independent networks*, N_{NW}** , the number of sub-networks consisting of linked paths between the connected streams:

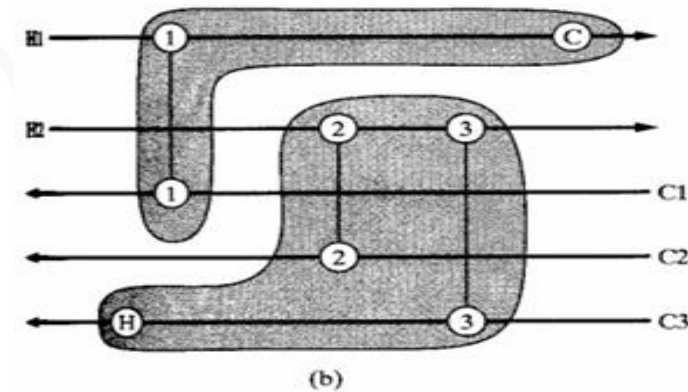
$$N_{HX,min} = N_S + N_U - N_{NW}$$



All streams in a process are connected directly or indirectly by heat exchangers

$$N_{NW} = 1, N_{HX,min} = 3 + 2 - 1 = 4$$

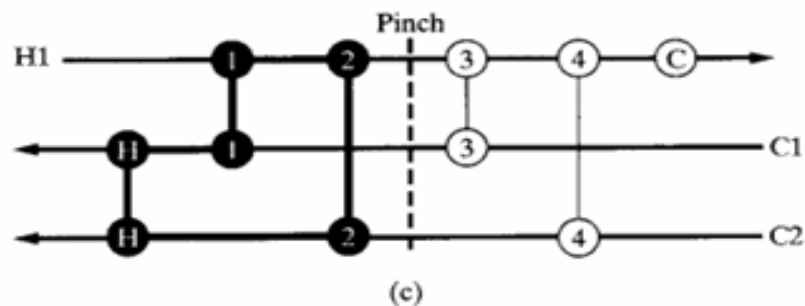
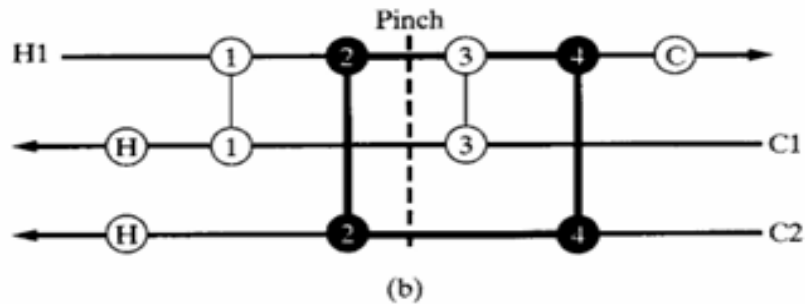
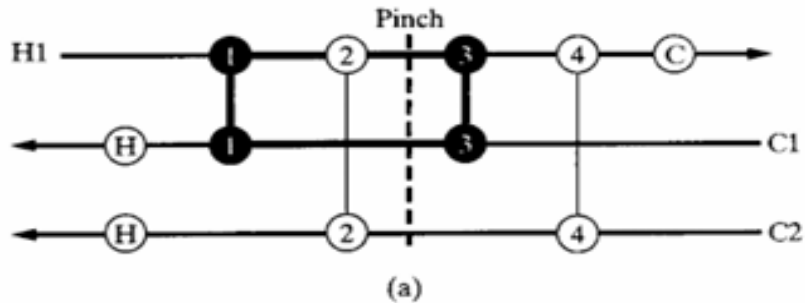
However, 7 HEs present, 3 more than min.



Five stream, 2 utilities, 2 independent subsystems: one connecting streams H1, C1 and cold utility, and second connecting streams H2, C2, C3 & hot utility.

$$N_{HX,min} = 5 + 2 - 2 = 5$$

Reducing the Number of Heat Exchangers- Breaking Heat Loops



- Hohmann (1971) showed that in a HEN with N_{HX} heat exchangers, $N_{HX} - N_{HX,min}$ independent heat loops exist - subnetworks that exhibit cyclic heat flows between two or more streams.
- Simplest case of heat loops in Figure a and b, streams H1 and C1 are matched twice with heat exchangers 1 and 3; heat flows between streams H1 and C2 in two matches, in heat exchangers 2 and 4.
- A more complex heat loop is shown in Figure c, in which heat flows between streams H1 and C1 in heat exchanger 1, between H1 and C2 in heat exchanger 2, and between C1 and C2 and the hot utility, forming a closed heat cycle.
- The three heat loops identified in explain why the HEN has three heat exchangers more that $N_{HX,min}$.

Reducing the Number of Heat Exchangers- Breaking Heat Loops

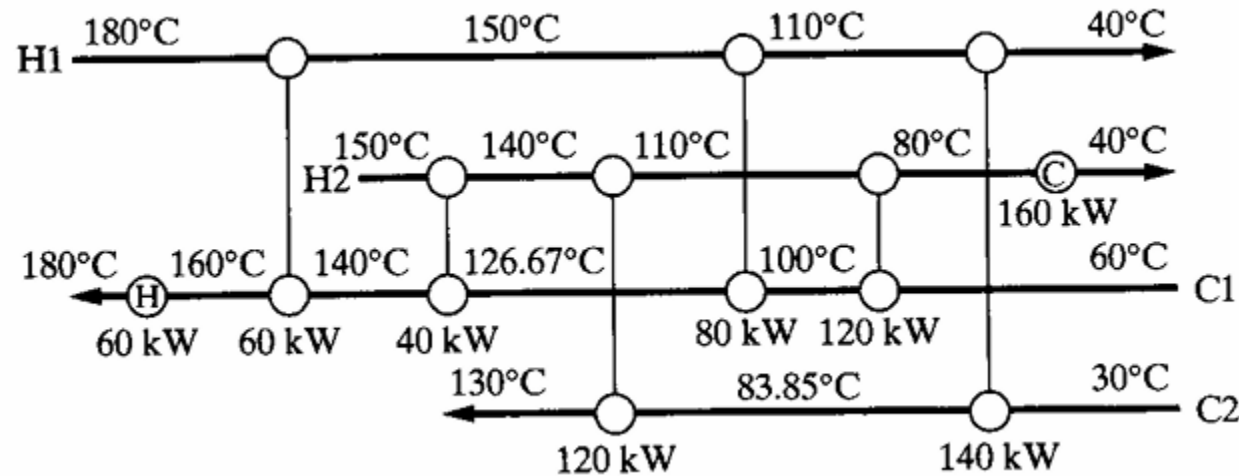
- Since a pinch exists in Figure, a HEN that meets the MER targets consists of two independent networks, one on each side of the pinch, so that

$$N_{HX,min}^{+} = N_{HX,min}^{-} = N_S + N_U - 1 = 3 + 1 - 1 = 3$$

- The '+' and '-' superscripts indicate the hot and cold side of the pinch.
- **Thus, the HEN in Figure can be designed to meet the MER targets with just six heat exchangers, ($N_{HX,min}^{MER} = N_{HX,min}^{+} + N_{HX,min}^{-} = 6$).**
- The simplest change is to eliminate heat exchanger 1, transferring its duty to heat exchanger 2, decreasing the duty of the heater on stream C2 by this amount, and increasing the duty of the heater on stream C1 by this amount. This is referred to as **loop breaking**.
- Often, **the smallest heat exchanger in the heat loop is eliminated**, because the cost of the area saved by eliminating a small exchanger is usually more than the cost incurred in increasing the area of a large exchanger by the same amount.
- **Each heat loop is broken by removing a heat exchanger and adjusting the heat loads accordingly, which often leads to heat flow across the pinch, moving the HEN away from its MER targets.**

Example 10.9

Return to the HEN in Figure 10.15d, which was designed to meet its MER targets, and involves eight heat exchangers, three more than $N_{HX,min}$ given by Eq. (10.13). In this example, the heat loops are identified and removed, one by one, taking note of the impact on the capital cost, the cost of the utilities, and the annualized cost, C_A , in Eq. (10.3).



$$C_A = i_m \left[\sum_i C_{P,I_i} + \sum_j C_{P,A_j} \right] + sF_s + (cw)F_{cw}$$

Cooling water (cw): $T^s = 30^\circ\text{C}$, $T^r \leq 80^\circ\text{C}$, cost of cw = 0.00015 \$/kg

Steam (sat'd., s): $T = 258^\circ\text{C}$, $\Delta H^v = 1,676 \text{ kJ/kg}$, cost of s = 0.006 \$/kg

Overall heat-transfer coefficients:

$$U_{\text{heater}} = 1 \text{ kW/m}^2 \text{ }^\circ\text{C}, U_{\text{cooler}} = U_{\text{exch}} = 0.75 \text{ kW/m}^2 \text{ }^\circ\text{C}$$

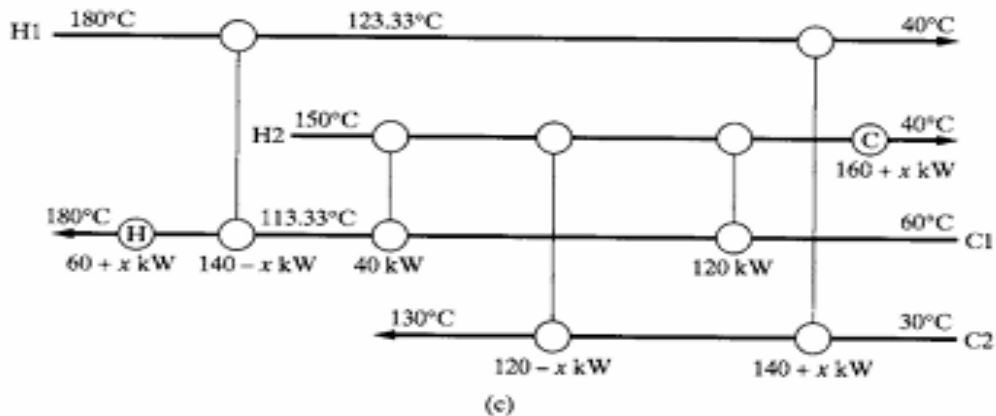
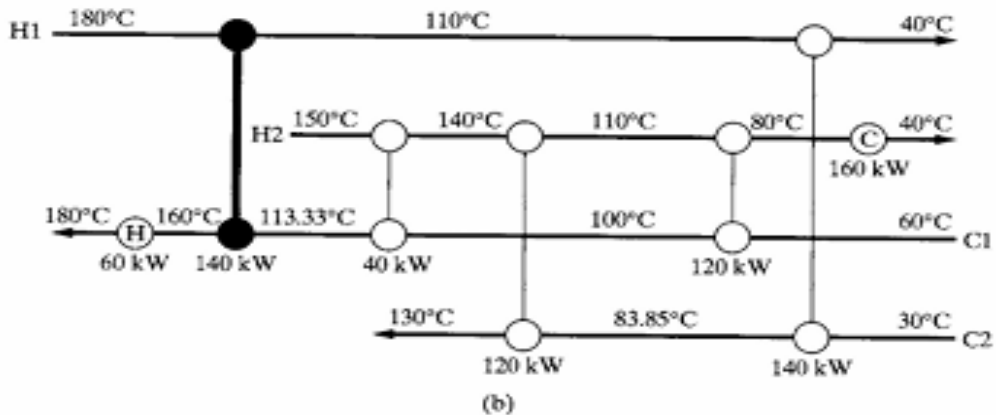
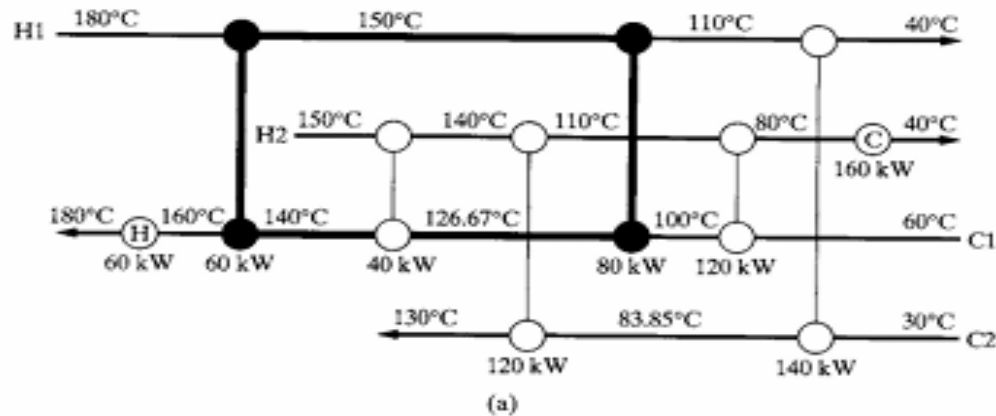
Purchase cost of heat exchangers:

$$C_p = 3,000A^{0.5} \quad (\$, \text{ m}^2)$$

Equipment operability = 8,500 hr/yr, return on investment = $i_m = 0.1$ [for Eq.(10.3)].

$$A = Q/(UF_T\Delta T_{LM})$$

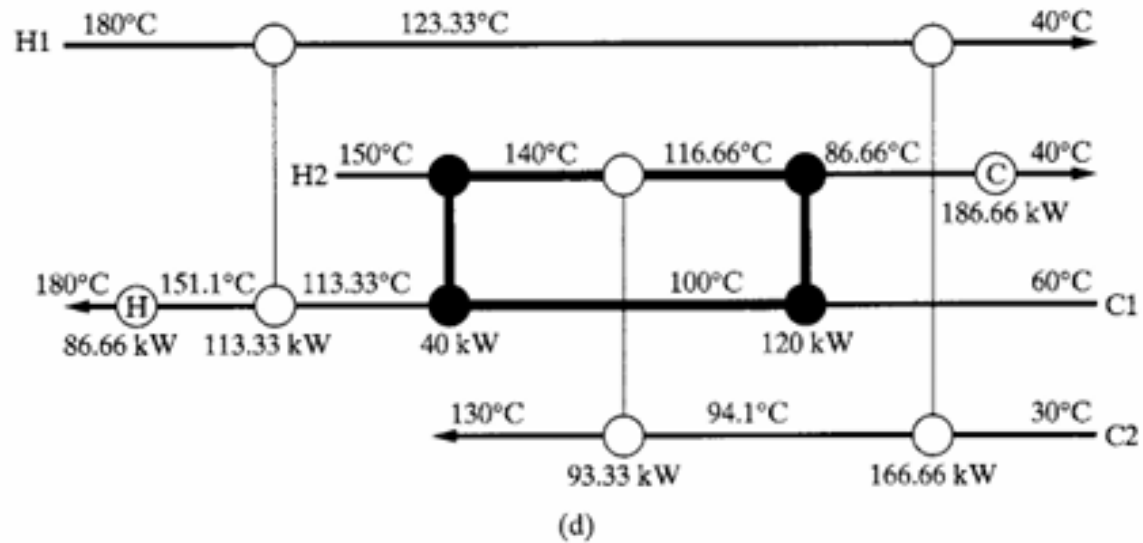
- Beginning with the HEN in Figure d, the heat transfer area for each heat exchanger is computed.
- Purchase costs are estimated using $C_p = 3000A^{0.5}$, and the total purchase cost is computed to be \$66,900.
- The annual cost of steam and cooling water is \$10,960/yr, which combines with the purchase cost, multiplied by a return on investment ($i_m = 0.1$), to give $C_A = \$17,650/\text{yr}$.



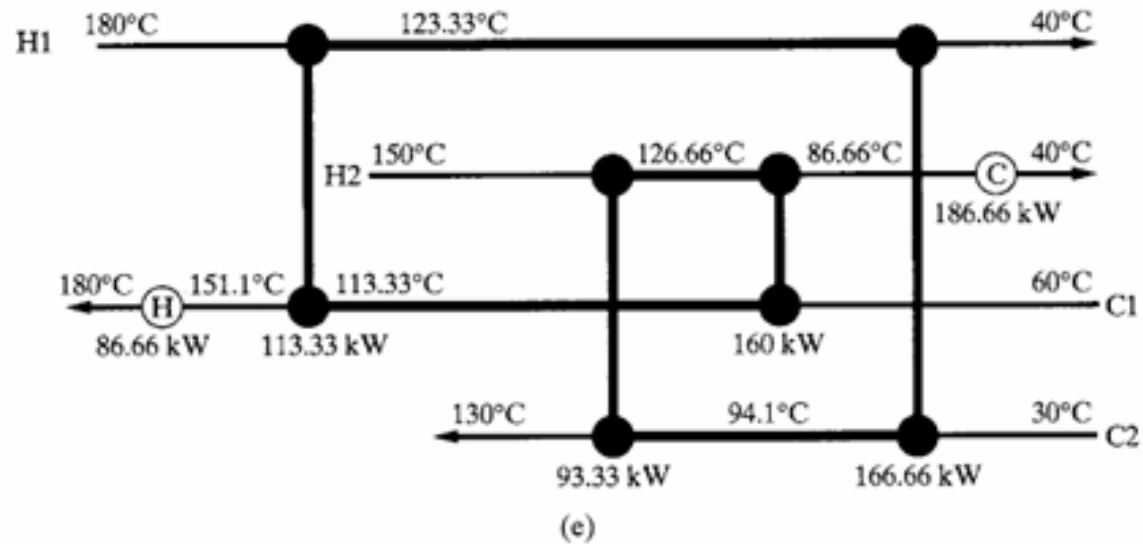
- Total: 8 heat exchangers, 1st loop between HES on H1 and C1
- 80 kW HE combined with 60 kW HE

- Seven heat exchangers, $\Delta T_{\min}$ violation
- Causes temperature approach violation (110-113.33 °C = -13.33 °C)
- Solved by transferring less heat from HE
- Amount computed to give $\Delta T_{\min} = 10$ °C on cold side, thus temp. of stream H1 reduced to 123.33 °C

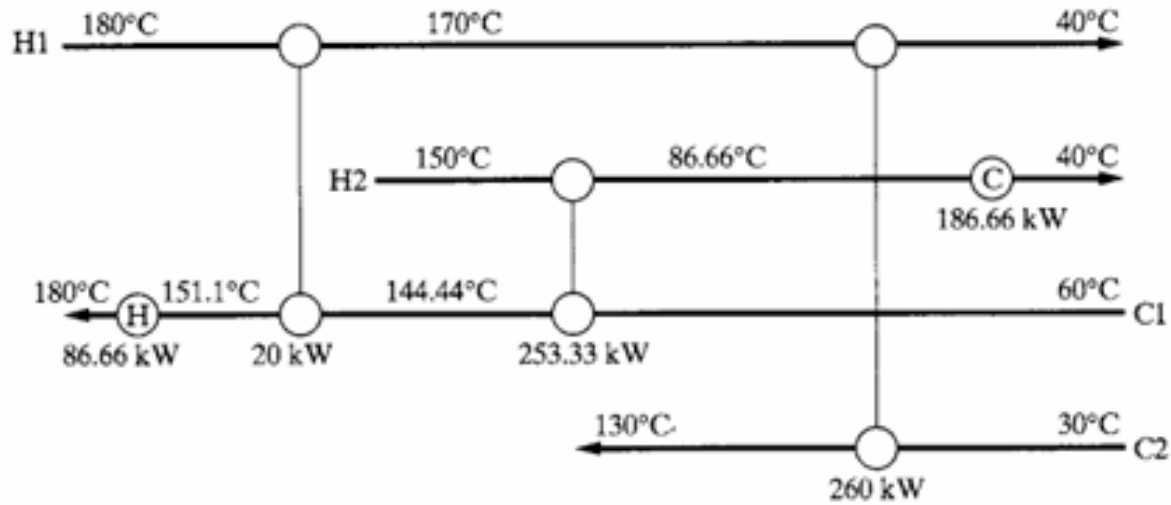
- Seven heat exchangers after shifting heat load
- By heat balance, $140 - x = 2(180 - 123.33)$
- Reduction in heat load $x = 26.66$ kW



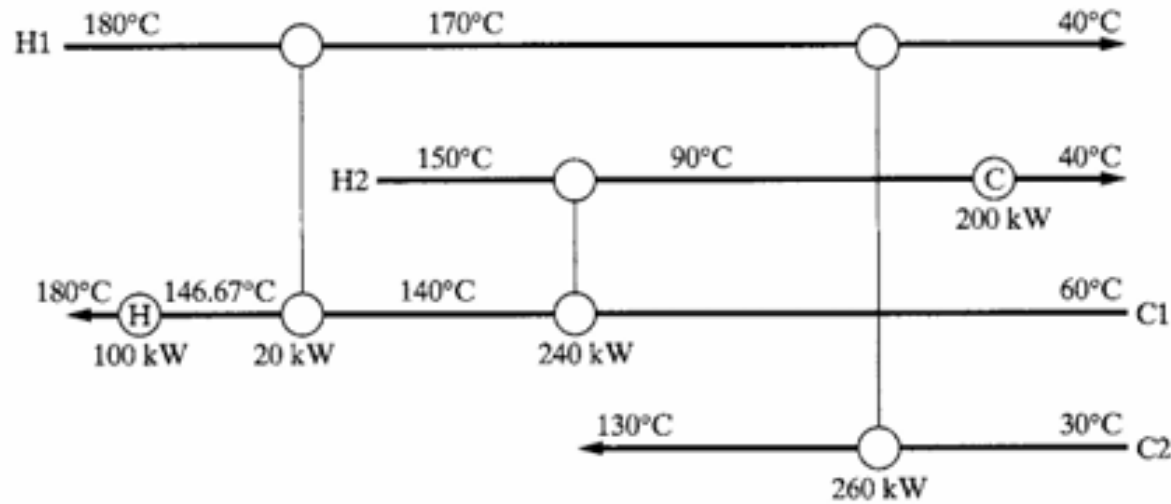
- Seven HE
- Second loop between H2 and C1
- 40 kW HE combined with 120 kW HE
- No temperature approach violation
- No adjustment in duty



- Six HE, third heat loop by all streams
- Eliminated by shifting the load of the smallest HE around heat loop



(f)



(g)

- Temperature approach violation = $150 - 144.44 = 5.56 \leq 10^\circ\text{C}$
- 253.33 kW heat load is reduced by y
- $253.33 - y = 3$ (14-60)
- $y = 13.33$ kW
- Enables stream C1 to leave HE at 140°C

- $y = 13.33$ kW added and removed by additional steam and cooling water
- **Five heat exchangers**

Table 10.5 Cost Comparison for Example 10.9

Network	Utilities Cost (\$/yr)	C_P , Purchase Cost (\$)	C_A , Annualized Cost (\$/yr)
Design for MER: 8 HXs	10,960	66,900	17,650
7 HXs	13,250	57,470	19,000
6 HXs	13,250	54,430	18,690
Design for $N_{HX,min}$: 5 HXs	15,340	45,930	19,930
Design for MER: stream- split design, 6 HXs ^a	10,960	54,890	16,450

^aSee Example 10.11 for details of the stream-split design to meet $N_{HX,min}^{MER}$

- Trade-off between the total purchase cost and the cost of utilities shifts as the cost of fuel decreases and the capital cost or the return on investment increases

Stream Splitting

- When designing a HEN to meet its MER targets, **stream splitting** must be employed if:
 - (a) the number of hot streams at the pinch, on the cold side, is less than the number of cold streams; or
 - (b) the number of cold streams at the pinch, on the hot side, is less than the number of hot streams.
- In this way, parallel pairings that fully exploit the temperature differences between energy sources and sinks are possible.
- **Moreover, stream splitting helps to reduce the number of heat exchangers in a HEN without increasing the utility duties, which often occurs when heat loops are broken.**
- Stream splitting is also used to reduce the number of heat exchangers for HENs that satisfy the MER targets.
- While stream splitting provides these advantages, its use should be considered with **caution because it reduces flexibility and complicates process operability.**
- When possible, it is usually **preferable to break heat loops without stream splitting,**

Generalized rules for stream splitting on both sides of the pinch to satisfy MER requirements

Hot Side of Pinch:

H-1. Let N_H and N_C be the number of hot and cold streams at the pinch. Noting that cold utilities cannot be used on the hot side of the pinch, if $N_H > N_C$, a cold stream must be split, since for feasibility, $N_H \leq N_C$.

H-2. On the hot side of the pinch, all feasible matches must ensure $C_h \leq C_c$. If this is not possible for every match, split one or more hot streams as necessary. If hot streams are split, return to step H-1 above.

Cold Side of Pinch:

C-1. Let N_H and N_C be the number of hot and cold streams at the pinch. Noting that hot utilities cannot be used on the cold side of the pinch, if $N_C > N_H$, a hot stream must be split, since for feasibility, $N_C \leq N_H$.

C-2. On the cold side of the pinch, all feasible matches must ensure $C_c \leq C_h$. If this is not possible for every match, split one or more cold streams as necessary. If cold streams are split, return to step C-1 above.

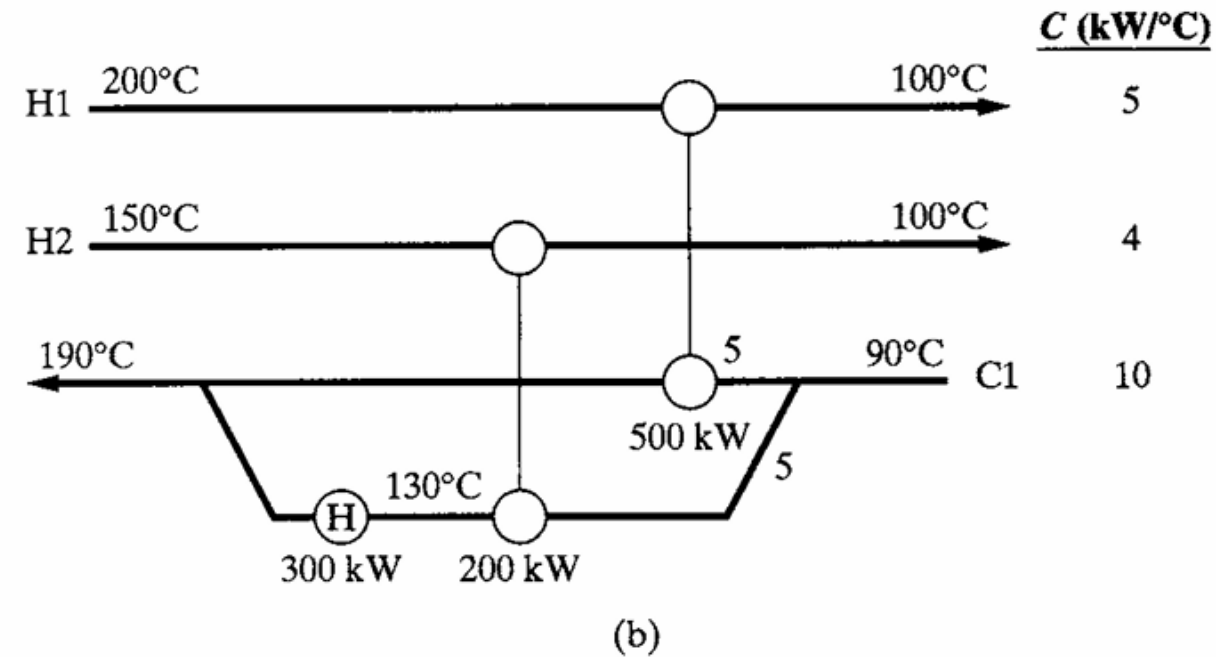
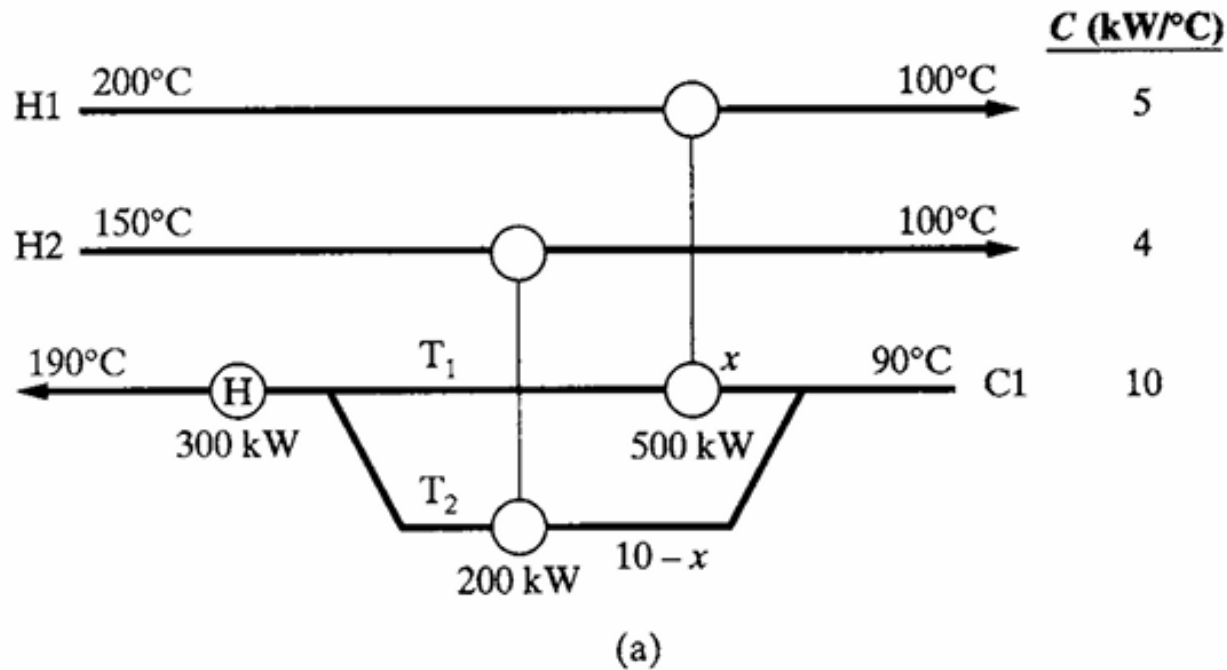
Example 10.10

It is required to design a HEN, with a minimum number of heat exchangers that satisfy $\Delta T_{\min} = 10^\circ\text{C}$ and a hot utility MER target of 300 kW, for three streams on the hot side of the pinch:

Stream	T^s ($^\circ\text{C}$)	T' ($^\circ\text{C}$)	C (kW/ $^\circ\text{C}$)	Q (kW)
H1	200	100	5	500
H2	150	100	4	200
C1	90	190	10	1,000

- Since no cold utility is allowed, $N_{\text{HX},\min} = 3$.
- Furthermore, because the two hot streams need to be cooled entirely by the cold stream, to avoid an approach-temperature violation ($\Delta T_1 < \Delta T_{\min}$), the cold stream must be split.
- Note that the duties of all three heat exchangers have been set to satisfy the MER targets, but the portion of the heat-capacity flow rate assigned to the first stream, x , must be determined by solving the energy balances for the split streams.

- Note that by making a minor structural change in which the heater is moved upstream of the mixing junction, as in Figure 10.22b, the design is improved to provide isothermal mixing, as desired.



MER design for Example 10.10: (a) split determination; (b) improved design.

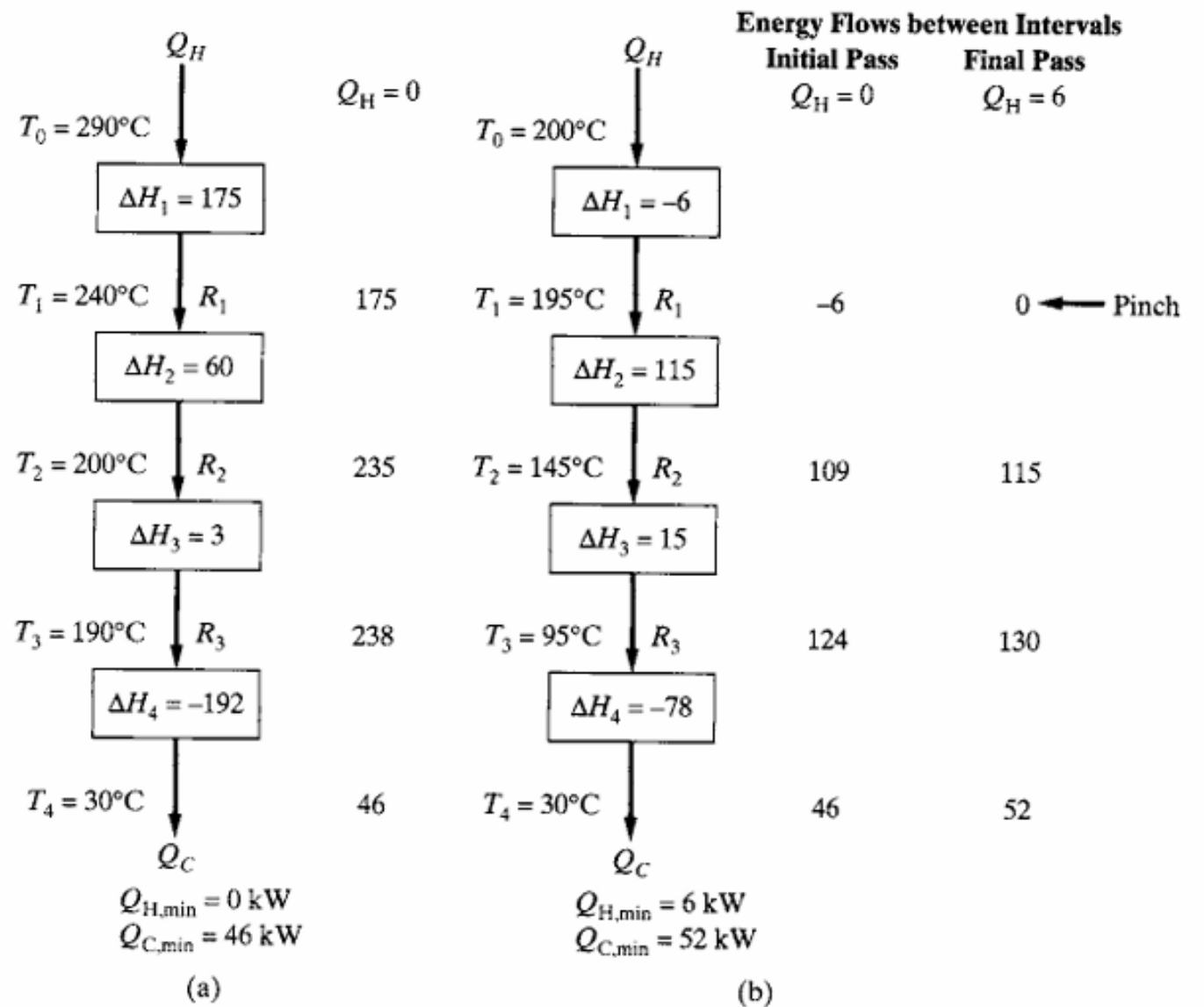
Threshold Approach Temperature

- In many cases, the selected $\Delta T_{\min}$ is such that no pinch exist, and MER design calls for either hot or cold utility to be used, but not both.
- **The critical $\Delta T_{\min}$ below which no pinch exists is referred to as the threshold approach temperature difference, Δ_{thres} .**

- **Example 10.12**

Compute MER targets for the following streams as a function of the minimum approach temperature:

Stream	T^s (°C)	T^t (°C)	C (kW/°C)	Q (kW)
H1	300	200	1.5	150
H2	300	250	2	100
C1	30	200	1.2	204



- For $\Delta T_{\min} = 10^\circ\text{C}$, no pinch exists, as shown in Figure 10.24a using the TI method.
- Note that 46 kW of cold utility are required.
- When the analysis is repeated as $\Delta T_{\min}$ is varied, for $\Delta T_{\min} > 100^\circ\text{C}$, a pinch exists, as illustrated in Figure 10.24b for $\Delta T_{\min} = 105^\circ\text{C}$.

Figure 10.24 Temperature intervals, energy balances, and residuals for Example 10.12 at: (a) $\Delta T_{\min} = 10^\circ\text{C}$; (b) $\Delta T_{\min} = 105^\circ\text{C}$.

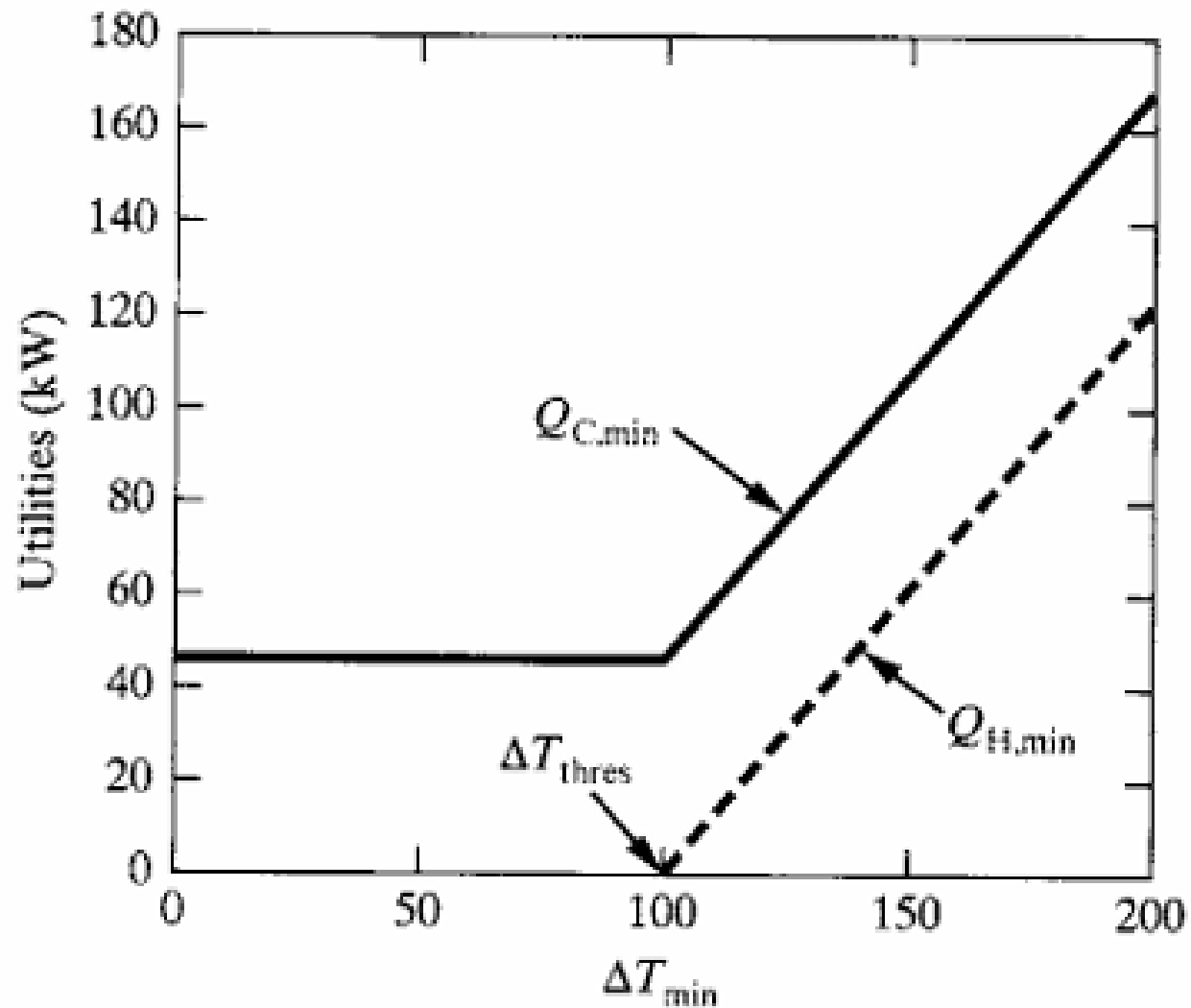


Figure 10.25 Minimum utility requirements as a function of $\Delta T_{\min}$ for Example 10.12.

- Figure 10.25 shows the threshold, $\Delta T_{\min} = \Delta T_{\text{thres}}$ at which the pinch appears.
- **Since no heating utility is required for $\Delta T_{\min} < \Delta T_{\text{thres}}$** where the cooling requirement is constant at 46 kW, no energy is saved while capital costs are increased as $\Delta T_{\min}$ is decreased.

Optimum Approach Temperature

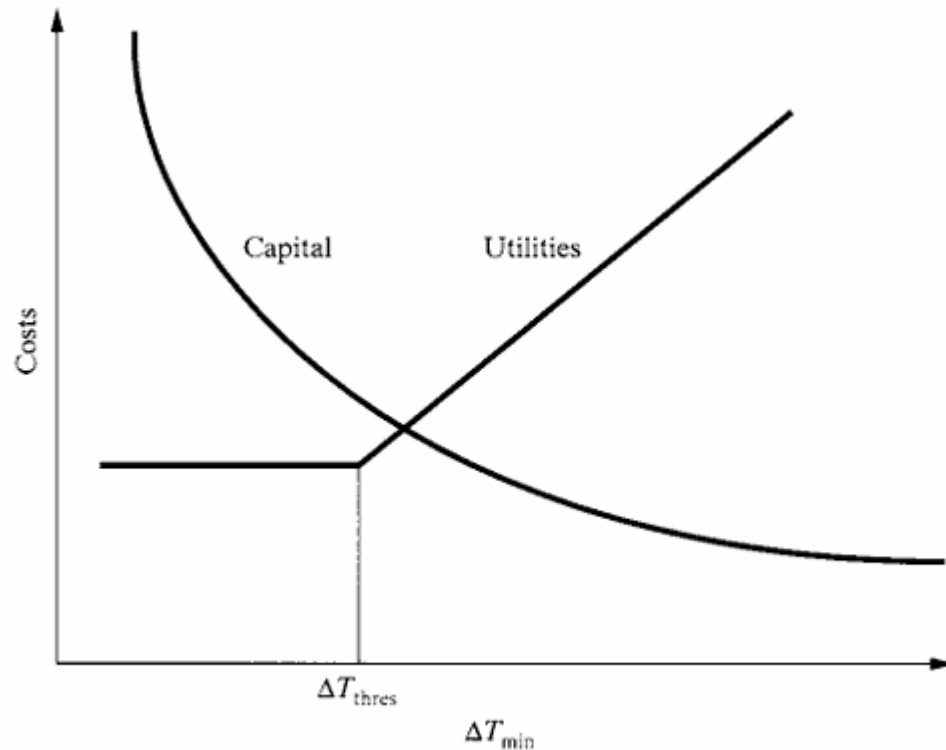
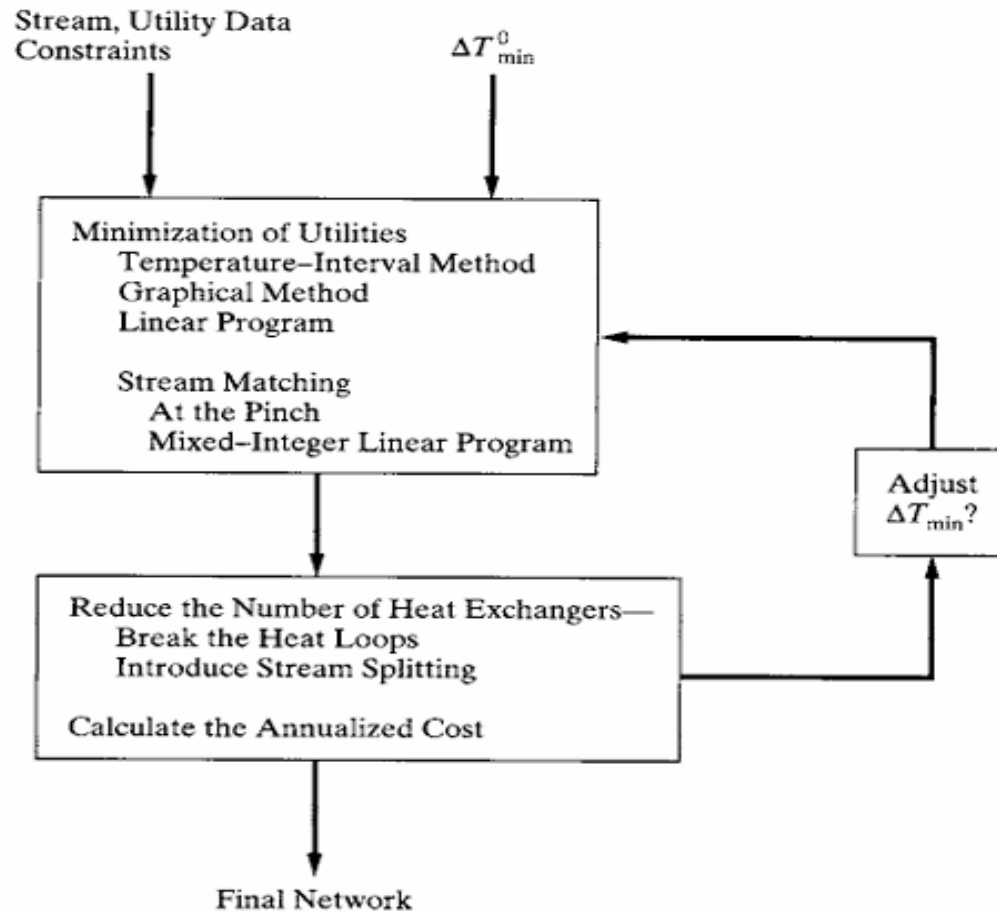


Figure 10.27 Trade-off between capital and utilities costs as a function of ΔT_{min} .

- As $\Delta T_{min} \rightarrow 0$, the true pinch is approached at which the area for heat transfer approaches infinity, while the utility requirements are reduced to the absolute minimum.
- At the other extreme, as $\Delta T_{min} \rightarrow \infty$, the heat transfer area approaches zero and the utility requirements are increased to the maximum, with no heat exchange between the hot and cold streams.
- The variations in heat transfer area and utility requirements with ΔT_{min} translate into the variations in capital and operating costs.
- *As ΔT_{min} decreases, the cost of utilities decreases linearly until a threshold temperature, ΔT_{thres} is reached, below which the cost of utilities is not reduced.*
- Thus, when $\Delta T_{min} \leq \Delta T_{thres}$ the trade-offs between the capital and utility costs do not apply.

Derivation of Network Superstructures for Minimization of Annual Costs



- Two-step procedure in which HENs are designed initially to have the minimum utilities, followed by a reduction in the number of heat exchangers.
- This strategy is particularly effective when the cost of fuel is high relative to the purchase costs of the heat exchangers.
- **When solved formally as optimization problems, given ΔT_{min} , the determination of the minimum utilities involves the solution of a linear programming (LP) problem.**
- The determination of the minimum number of matches involves the solution of the transshipment problem in the form of a mixed-integer linear program (MILP).

Derivation of Network Superstructures for Minimization of Annual Costs

- The second step of the process involves breaking the heat loops and stream splitting.
- This can be followed by the systematic adjustment of $\Delta T_{\min}$ to obtain HENs that have annualized costs closer to the global optimum.
- The overall strategy, tends to be less effective when the cost of fuel is low relative to the purchase costs of the heat exchangers.
- Networks having minimum utilities and a large number of heat exchangers are far from optimal.
- In their most general form, the formulations are mixed-integer nonlinear programs (MINLPs), involving nonlinear terms in the objective function and in the constraints.
- While the MINLPs can be formulated rather easily, unfortunately, the solution algorithms available today, in systems such as GAMS, are limited in their ability to obtain converged solutions.
- As computational speeds increase, along with the size of computer memories, algorithms that are more effective are being developed and more complex and larger mathematical programs are being solved.

Quick Recap

- What is Heat Exchanger Network (HEN) synthesis?
- What are the factors in cost analysis of HEN?
- What are the methods for MER targeting – minimum utilities?
- Write the steps for temperature interval method.
- What information is available for Hochmann/Lochart Composite(HCC) Curves?
- What are the advantages of Grand Composite Curves (GCC)?
- Write the steps to design HEN by stream matching at pinch.
- Describe the methods to obtain minimum number of heat exchangers in HEN.
- What is threshold and optimum approach temperature?
- What is the purpose of network superstructure?